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Privacy-Aware Federated Recommender Systems for IoT Healthcare: A Review of Computational Methods, Multimodality, and Optimization for Real-Time Personalization

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ABSTRACT

The rapid increase in the number of internet of things (IoT) devices in the healthcare sector has produced an ecosystem of multimodal data that includes physiological signals, behavioral measurements, and electronic health records. Nevertheless, centralized analytics are dangerous to the privacy and safety of patient data. Federated learning (FL) has become a prospective paradigm of decentralized intelligence, which allows the joint training of models without providing actual data. The review is a systematic review of research published during the 2020–2025 period on privacy-preserving federated recommender systems for healthcare applications in IoT. With the PRISMA protocol, the studies of interest are categorized under five thematic aspects: (i) privacy preservation techniques including differential privacy, secure aggregation, and homomorphic encryption, (ii) personalization and model optimization approaches, (iii) multimodal fusion systems, (iv) communication efficiency systems, and (v) metrics of evaluation in real-time implementation. As the analysis indicates, hybrid privacy models and adaptive optimization can make a major contribution to the improvement of personalization and scalability, but the issue remains in the manner of providing efficient communication and fairness between heterogeneous clients. In addition, the paper identifies open research directions, including quantum-resilient security and the integration of edge intelligence and sustainability-monitoring model training. The results of this review provide a single taxonomy, benchmarking standards, and a research direction for future studies to improve trustworthy, real-time, and privacy-compliant recommender systems in healthcare IoT settings.

1 | Introduction

Recent papers prove that federated learning (FL) can make healthcare modeling collaborative and does not centralize patient data [1–3]. The recent proliferation of internet of things (IoT) devices, wearable sensors, and electronic health records (EHRs) has generated extensive amounts of heterogeneous data at hospitals, clinics, and personal devices, as shown in Figure 1. These decentralized sources record real-time physiological, behavioral, and contextual signals that provide new opportunities for predictive and individualized healthcare services. However, conventional centralized learning pipelines continue to use the approach of aggregate sensitive data in cloud repositories, which are both privacy compliant and latency-sensitive. Ethics like regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR) also pose limitations on mass sharing of patient information that limit traditional model creation. Meanwhile, personalized healthcare needs a model that is sensitive to individual physiological and lifestyle variations, though convergence of models is challenged when heterogeneous data are involved, label imbalance, and even device variability. These limitations drive decentralized intelligence as one of the main paradigms of privacy-sensitive analytics and cross-institutional cooperation in the absence of raw data sharing [4–6].

1.1 | Motivation

FL has emerged as a highly popular decentralized method that allows the model training process to be trained on distributed nodes with data being localized. FL can foster the learning process in healthcare between various institutions or IoT devices, without breaking the principles of data privacy. However, when it comes to actual implementation, there is a threefold task: to provide the accuracy of the model, to guarantee the preservation of privacy, and to ensure the efficiency of communication [7–10]. A large value of differential privacy (DP) tends to spam model performance, whereas frequent parameter updates on low-bandwidth networks cause a substantial communication overhead. On the same note, secure aggregation (SA) and encryption systems, though paramount to confidentiality, increase the complexity of computations. Therefore, striking a balance between these three conflicting ends is still a research endeavor. The solution to this balance is scalable, real-time healthcare recommender systems that could assist in providing constant monitoring, risk forecasting, and adaptive treatment planning [11–13].

1.2 | Rationale and Contributions of This Study

The fast pace of FL and privacy-aware artificial intelligence has given rise to new possibilities of collaborative healthcare analytics that do not interfere with patient privacy. The current FL literature in healthcare is, however, disjointed in the areas of data privacy, multimodal learning, personalization, and optimization of communication [14]. Numerous of the contributions focus on a single aspect, including improving privacy or personalization without examining the interactions between the two, to enable real-time, patient-centered recommendation systems. Besides this, the lack of common taxonomies, benchmarking systems, and repeatable performance metrics impedes cross-study comparison and delays translation into systems to be put to use.

To overcome these drawbacks, this research paper conducts an integrative review that is structured, systematic, and guided. The detailed contributions are as follows:

1. Comprehensive systematic review: Adopts a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)−based approach to perform a systematic review of the literature published within the 2020–2025 timeframe on the topics of FL, privacy-sensitive recommender systems, and the IoT-oriented healthcare architectures.
2. Unified multidimensional taxonomy: This suggests a five-dimensional taxonomy of privacy preservation mechanisms, personalization techniques, multimodal data fusion strategies, communication efficiency models, and optimization strategies and allows one to analyze the current research trends in a unified manner.
3. Cross-domain benchmarking framework: Provides a comparative framework between healthcare datasets, privacy budgets (e), performance metrics (AUROC/F1), and communication overheads, supporting the reproducible evaluation of heterogeneous FL settings.
4. Critical insights and research gaps: Uncovers the unresolved problems including the inconsistent quantification of privacy, the lack of multimodal validation, the poor processing of non-IID data, and the absence of communication-accuracy trade-off studies in healthcare recommenders.
5. Future research directions: Details the future directions such as quantum-resilient security models, hybrid edge-cloud collaboration, energy-efficient training paradigms, and sustainable AI frameworks of the next-generation privacy-conscious healthcare recommendation systems.

By such contributions, the review offers a conceptual framework as well as empirical guidelines towards the attainment of credible, adaptable, and communication-effective federated recommender systems in decentralized IoT-based healthcare ecosystems.

2 | Background and Conceptual Foundations

2.1 | FL Architectures in Healthcare

FL has the capability to provide many healthcare organizations, IoT hardware, and hospital systems with the ability to jointly train models without sharing raw patient data. Rather than concentrating sensitive datasets, model updates are performed at local nodes and aggregated across the entire globe on a coordinating server, which sacrifices knowledge sharing but maintains privacy of the information in decentralized environments [15, 16]. In the medical context, FL architecture is mainly categorized into cross-silo, cross-device, and hybrid (multitier) architecture. The cross-device FL in Figure 2 links many wearables or sensors that have intermittent connectivity. The cross-silo FL in Figure 3 is used to connect institutions, including hospitals or research laboratories. Figure 4 combines the two in the hybrid approach, allowing cloud-level optimization and edge-level personalization [17].

FedAvg, FedProx, and FedNova are core optimization algorithms that are modified to deal with heterogeneous data distribution and device heterogeneity, as shown in Figure 5, 6, and 7, respectively [18]. These foundations are further elaborated with advanced frameworks that are coordinated by cloud and edge-end controls and privacy-saving facilities that adhere to medical data governance policies [19, 20]. Studies confirm that FL can achieve near-centralized accuracy in diagnostic prediction and

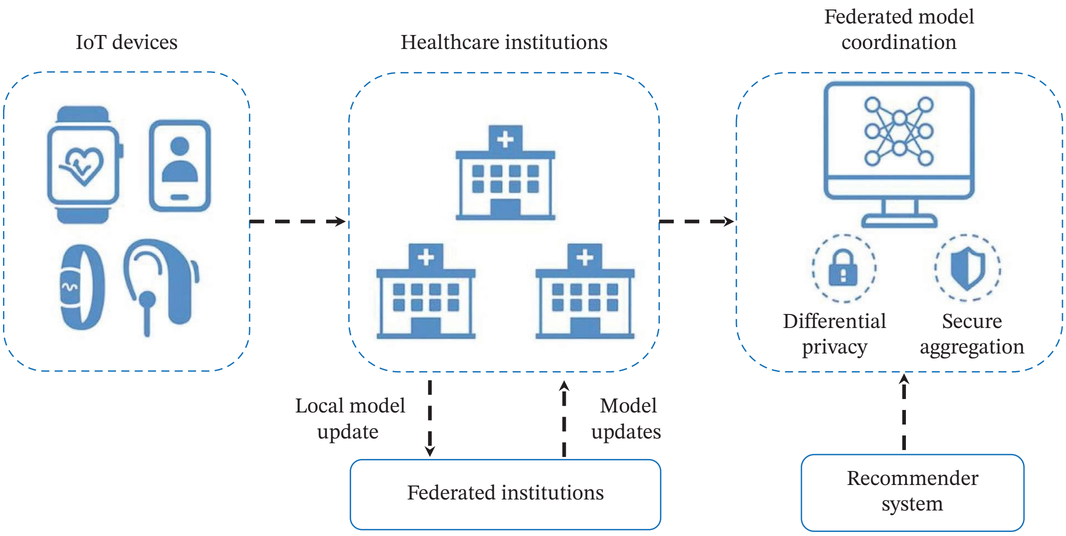

FIGURE 1 | Overview of privacy-aware federated healthcare recommender system. This schematic represents the decentralized IoT devices and healthcare institutions involved in FL under privacy-preserving schemes of differential privacy and secure aggregation coordinated by a central model server providing customized healthcare prescriptions.

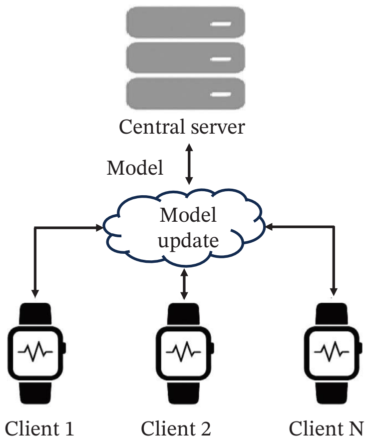

FIGURE 2 | Cross-device FL architecture. This figure shows various IoT healthcare devices (clients) participating in model training together with a central server. Patient information is stored on each device, and only model updates are sent so as to make learning decentralized and privacy-protecting.

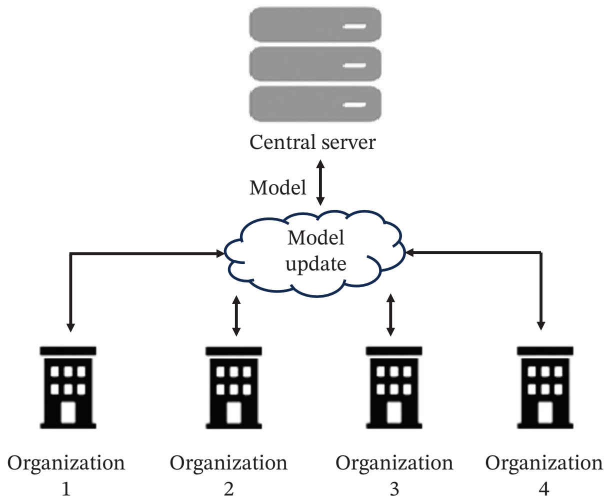

FIGURE 3 | Cross-silo FL architecture. This figure illustrates that several healthcare establishments serve as data silos, and all the local models are trained on institutional datasets. Encrypted model updates are aggregated on a central server, allowing collaborative learning in different hospitals at the same time, but allows patient data to be safely stored in each individual hospital.

clinical imaging tasks while maintaining strict data confidentiality [21, 22].

As shown in Figure 8, the decentralized data sources (EHRs, medical images, and IoMT devices) remain within hospitals and edge devices, which act as federated clients in cross-device, cross-silo, or hybrid configurations. A central federation server orchestrates global model training using algorithms such as FedAvg, FedProx, and FedNova, while privacy-enhancing mechanisms (DP, SA, homomorphic encryption [HE], trusted execution environments [TEEs], and blockchain) protect model updates. The resulting global model supports downstream clinical decision support and recommender systems, evaluated using accuracy, AUC, privacy budget (ε), and communication/latency metrics.

2.2 | Healthcare IoT Ecosystems and Data Modalities

Healthcare IoT ecosystems generate continuous multimodal data, encompassing physiological, behavioral, environmental, and contextual signals. Common modalities include vital signs from biosensors (e.g., electrocardiogram [ECG], photoplethysmogram [PPG], electromyogram [EMG], and internet of medical things [IoMT]), movement data from accelerometers or gyroscopes, structured and unstructured EHRs, and audio or vocal biomarkers for neurological and respiratory assessment [23].

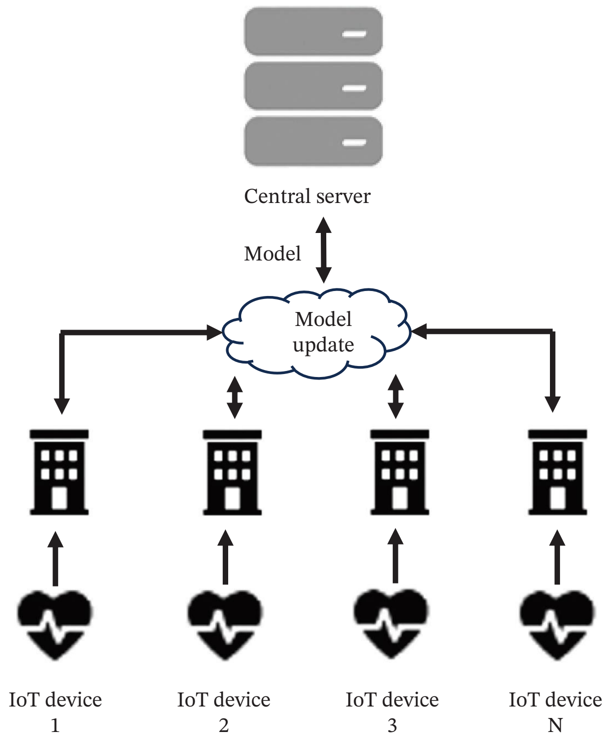

FIGURE 4 | Hybrid (multitier) FL architecture. The figure represents a hierarchical FL system in which the devices of the IoT healthcare system initially communicate with local hospital servers (edge tier), which in turn update the aggregated model to a global central server (cloud tier). The multilevel structure allows scalable and privacy-conserving learning between the devices and institutions in real-time.

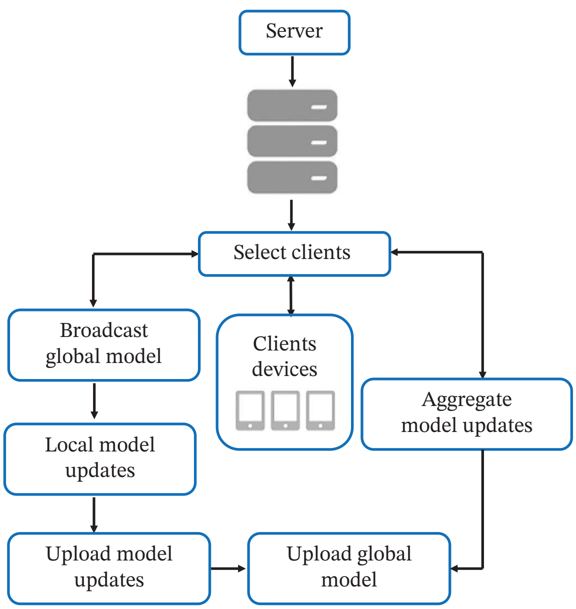

FIGURE 5 | Overview of the FedAvg (federated averaging) algorithm. In this, each selected client trains locally for a few epochs and sends model weights to the server, which averages them to form a new global model.

Integrating these modalities into unified learning frameworks remains challenging due to differences in sampling frequency, data sparsity, and quality across devices. Federated architectures must therefore support asynchronous updates, modality-specific encoders, and lightweight models for deployment on constrained edge devices [15, 19, 24].

Recent advances demonstrate successful multimodal fusion under privacy constraints, such as federated matrix factorization, transformer-based sequence encoders, and compressed representation sharing [25]. Moreover, distributed learning infrastructures now exploit cloud–edge synergy for efficient data aggregation and on-device personalization [16, 20, 26]. This evolution has made FL increasingly viable for large-scale healthcare IoT systems that must balance real-time analytics with data sovereignty [22].

2.3 | Privacy-Preserving Mechanisms

While FL avoids raw data transfer, privacy leakage may still occur through gradient inversion or membership-inference attacks. Hence, several privacy-enhancing technologies (PETs) have been integrated into healthcare FL pipelines [19].

2.3.1 | DP

DP injects calibrated noise into gradients to obscure individual contributions, safeguarding patient identity even against strong adversaries [19, 27]. A randomized algorithm $\mathcal{M}$ satisfies (ε, δ)-DP if for any two neighboring datasets D_1, D_2 differing in one record and for all possible outputs S

$$\Pr[\mathcal{M}(D_1) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(D_2) \in S] + \delta, \tag{1}$$

where $P =$ masking probability, $r =$ injected rating, $\varepsilon =$ privacy budget (smaller $\longrightarrow$ stronger privacy), and $\delta =$ probability of failure to achieve DP. The typical gradient perturbation in FL is

$$\widetilde{g} = g + \mathcal{N}\left(0, \sigma^2 C^2 I\right), \tag{2}$$

where g is the clipped gradient, C is the clipping norm, and σ controls noise scale.

2.3.2 | SA

SA employs cryptographic protocols to ensure that only the global sum of updates is visible to the server [20]. SA masks each client's update with random noise such that only the aggregate is revealed.

If client i sends $u_i + r_i$ (where r_i is a random vector) and pairwise masks cancel

$$\sum_i (u_i + r_i) - \sum_i r_i = \sum_i u_i, \tag{3}$$

then the server obtains only the total update $\sum_i u_i$, preserving individual privacy.

2.3.3 | HE

HE allows computations on encrypted data, though at a higher computational overhead [24].

HE allows arithmetic on ciphertexts such that decryption yields the same result as computing directly on plaintexts.

For additive HE (e.g., Paillier):

$$E(m_1) \cdot E(m_2) = E(m_1 + m_2), \tag{4}$$

$$E(m)^k = E(km). \tag{5}$$

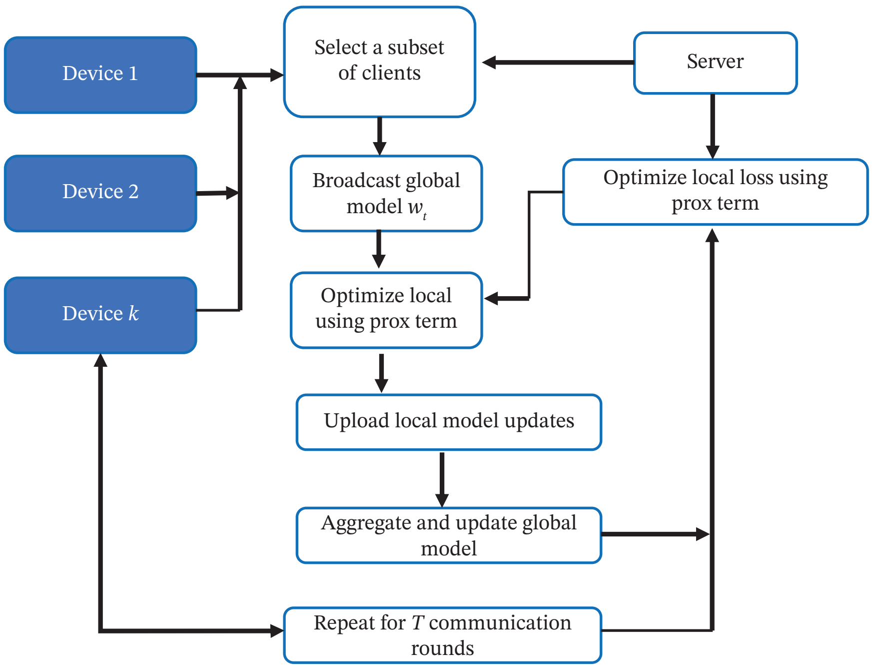

FIGURE 6 | Overview of FedProx (federated proximal) algorithm. It is an extension of FedAvg that adds a proximal term to the local loss so that each client's model does not drift too far from the global model, making training more stable under heterogeneous (non-IID) data and different computing capacities.

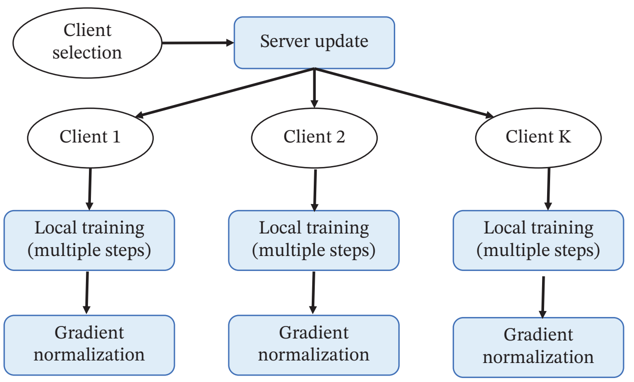

FIGURE 7 | Overview of FedNova (normalized averaging) algorithm. This normalizes client updates by their number of local steps and learning rates before aggregation, removing "objective inconsistency" and giving fairer, more stable convergence when clients train for different numbers of local iterations.

Thus, if clients encrypt model weights w_i, the server can compute

$$E\left(\sum_i w_i\right) = \prod_i E(w_i), \tag{6}$$

without ever seeing w_i in plaintext.

2.3.4 | TEEs

TEEs provide hardware-level isolation to protect data and computation integrity inside the device or edge node. A TEE (e.g., Intel SGX, ARM TrustZone, or AMD SEV) creates a secure enclave where sensitive model updates or aggregation steps can be executed in encrypted memory. Mathematically, computation f on data D_i satisfies

$$\mathrm{TEE}(f(D_i)) \Longrightarrow \mathrm{integrity}(f, D_i) = 1,\ \mathrm{confidentiality}(D_i) = \mathrm{preserved}. \tag{7}$$

That is, no external process including the system OS can tamper with or observe enclave data during computation. In healthcare FL, TEEs ensure that intermediate model gradients, cryptographic keys, and DP parameters remain shielded even from privileged attackers [13, 20, 28]. When TEE is used alongside DP or HE, the resulting "defense-in-depth" model can be achieved, including cryptographic protection and hardware isolation, complementing each other to provide a better level of data security.

2.3.5 | Blockchain Transparency

This includes an audit trail of immutable provenance of updates and accountability of models [29]. To improve trust and auditability of FL, blockchain supports an unalterable chain of updates of the model. The aggregate model parameters or hash summaries of each round are stored as a block with the cryptographic link to the preceding block.

2.3.5.1 | Block Structure Formula.

Each block B_t in round t contains

$$B_t = \left(H(B_{t-1}), \mathrm{Agg}(w_t), \tau_t, \mathrm{Sig}_{\mathrm{server}}\right), \tag{8}$$

where $H(B_{t-1})$ = hash of the previous block (ensures immutability), $\mathrm{Agg}(w_t)$ = aggregated global model weights at round t, τ_t = timestamp, and $\mathrm{Sig}_{\mathrm{server}}$ = digital signature verifying authenticity.

The blockchain ledger, therefore, satisfies

$$H(B_t) = \mathrm{SHA256}(B_t), \text{ and } H(B_t) \neq H'(B_t)\ \forall \text{ tampered } B_t. \tag{9}$$

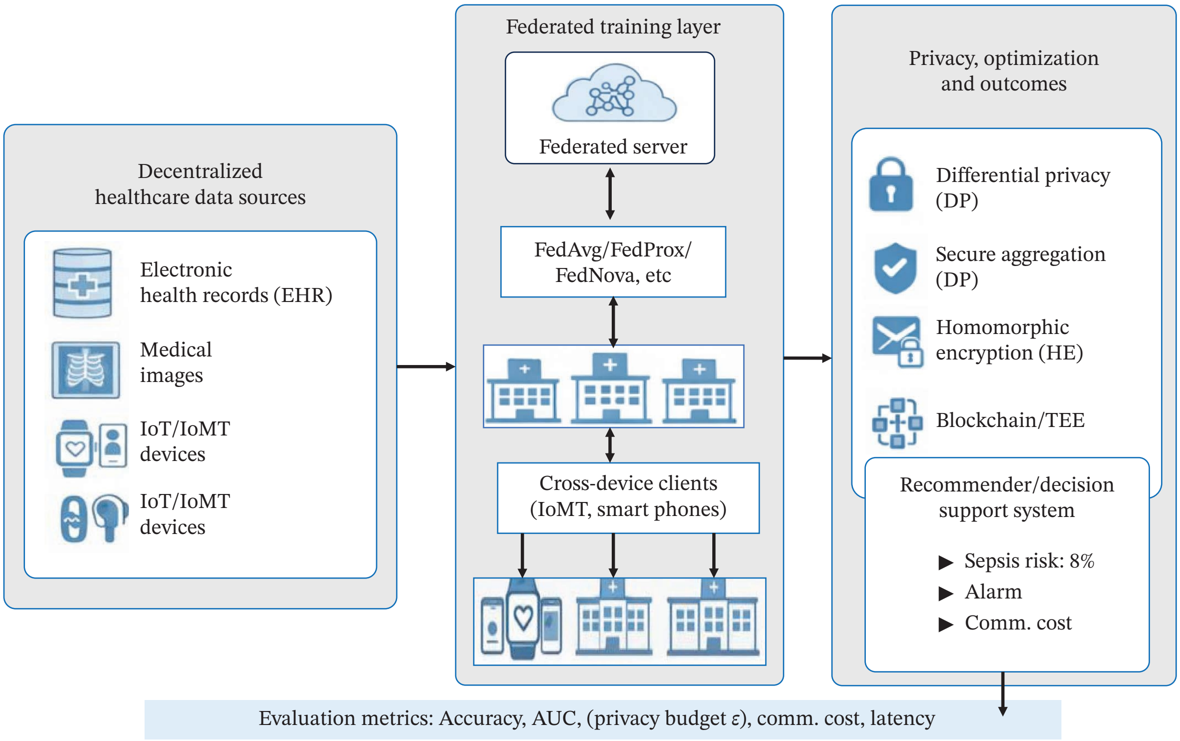

FIGURE 8 | Consolidated overview of the healthcare federated learning pipeline.

Any change of the model parameters of the past invalidates all further hashes, and it guarantees transparent model provenance among participants [29].

2.3.5.2 | Consensus Representation. To integrate trustless participation among institutions

$$\text{Consensus: } \mathrm{Agg}(w_t) = f_{\text{consensus}}\left(\left\{E\left(w_t^i\right)\right\}_{i=1}^N\right), \tag{10}$$

where $f_{\text{consensus}}$ may be the proof-of-authority (PoA) or practical Byzantine fault tolerance (PBFT), securing synchronization of global model updates on-chain [29].

2.3.6 | Federated Recommender Systems

Federated recommender systems also combine such PETs with collaborative filters and matrix factorization systems to provide privacy-sensitive personalization [17, 23, 25, 27]. Recent studies emphasize combining DP with SA or blockchain layers to balance privacy, traceability, and efficiency in IoT healthcare [21, 29, 30]. Collectively, these mechanisms underpin trustworthy, regulation-compliant federated systems aligned with HIPAA, GDPR, and other international standards [20]. Federated recommender systems extend collaborative filtering or neural recommendation models into a distributed, privacy-preserving setup. Each client (hospital, wearable device, or app) maintains its local user–item interaction matrix R_i and trains a local model f_i without exposing R_i.

2.3.6.1 | Local Optimization Objective. For client i, the local objective for matrix factorization–based FRS is

$$\min_{\{U_i,V_i\}} l_i = \sum_{\{(u,v)\in R_i\}} \left(R_{\{i,uv\}} - U_{\{i,u\}^{\{T\}V}_{\{i,v\}}}\right) + \lambda\left(\left|U_{i|}^{\{2\}}\right| + \left|V_{i|}^{\{2\}}\right|\right), \tag{11}$$

where U_i, V_i = latent feature vectors of users and items and λ = regularization parameter.

Each client computes gradients $\nabla \mathcal{L}_i$ locally and sends updates to the server.

2.3.6.2 | Federated Aggregation. The global model is updated as

$$U, V \longleftarrow \text{frac}\{1\}\{N\} \sum_{\{i=1\}}^{\{N\}} U_i, V_i, \tag{12}$$

or in weighted form (FedAvg style) as

$$U, V \longleftarrow \sum_{i=1}^{N} \{n_i\}\{n_{\{\{\text{total}\}\}}\}(U_i, V_i), \tag{13}$$

where n_i = number of interactions on client i.

In neural recommender variants (e.g., autoencoder- or transformer-based FRS), the shared model f_θ is optimized as

$$\theta_{\{t+1\}} = \sum_{\{i=1\}}^{\{N\}} \frac{n_i}{ntotal}\left(\theta_t^i - \eta\nabla_{\theta_t^i L_i}\right). \tag{14}$$

This enables decentralized personalization while preserving local data privacy [23, 25, 30].

2.4 | Open-Source FL Frameworks

Besides algorithmic contributions, various open-source frameworks have also become available to facilitate the design, simulation, and deployment of FL systems into practice. Table 1 provides an overview of typical frameworks and their key peculiarities with an emphasis on the features that are especially important in healthcare applications.

TABLE 1 | Widely used FL frameworks.

Framework	Primary language/stack	Key features	Healthcare relevance
Flower [11, 13, 31–33]	Python, framework-agnostic (PyTorch, TensorFlow, etc.)	Flexible client–server orchestration, strategy API, simulation, secure aggregation support	Commonly used to prototype healthcare FL with HE and multi-institution simulations
TFF [34–37]	Python/TensorFlow	Declarative federated computations, tight TF integration, simulation-oriented	Suited for academic EHR and medical imaging experiments in TensorFlow
FedML [12]	Python; supports PyTorch and TensorFlow	End-to-end FL pipeline, cross-device/cross-silo modes, experiment management, edge support	Enables benchmarking and deployment on edge/IoT healthcare platforms
OpenFL [38–40]	Python; Intel/enterprise-oriented	Secure enclaves, configurable topologies, hardware accelerator integration	Aimed at privacy-sensitive, production FL in regulated healthcare settings
PySyft [40–42]	Python	FL + MPC + DP on tensors, rich PET experimentation	Ideal for privacy-enhancing method research and experimental healthcare FL prototypes

3 | Review Methodology

3.1 | PRISMA-Based Selection (2020–2025)

Figure 9 shows the PRISMA approach, which shows how identification and screening are performed along with the included studies. Screening was conducted on the title/abstract and full-text using a standardized form by two reviewers independently. The first form of discussion was on the inclusion or exclusion disagreements, but the disagreements that could not be resolved (less than 5% of records) were resolved by a third senior reviewer. The decision taken at the end of the PRISMA flow diagram is then one that concurred with all reviewers.

The search limit was set to 1 January 2020 to 30 October 2025, since large-scale healthcare FL deployments and PET (e.g., DP, secure aggregation, and HE) implementations did not start to enter significant amounts before this time. This period also includes the post-COVID-19 changes, where the collaboration between institutions, regulatory policies, and digital health infrastructure benefited.

3.2 | Inclusion/Exclusion Criteria

The databases such as Scopus, IEEE Xplore, PubMed, WoS, ACM, arXiv, and others are explored from 1 January 2020 to 20 October 2025. The following search terms are used ("federated learning" OR "federated analytics" OR "federated recommender") AND (health OR "medical" OR "clinical" OR "electronic health record" OR EHR OR "medical imaging" OR "smart healthcare" OR "IoMT" OR "health IoT"). For this review, studies were included if they were in English, published or posted between 2020 and 2025, used FL or closely related decentralized learning as a core method, and focused on a healthcare or health-relevant setting (clinical prediction, medical imaging, EHR, wearables/IoMT, and health recommender systems). Eligible papers had to describe sufficient technical detail on at least one of the following: privacy/security mechanisms (DP, HE, SA, TEE, and blockchain), personalization or optimization strategies, multimodal fusion, or communication/scalability techniques, and report quantitative performance metrics (e.g., accuracy/AUC and, where available, ε, communication cost, or latency).

Excluded were nonhealth domains without transferable methods, generic AI/ethics commentaries, security or blockchain works with no FL, FL works lacking any concrete privacy/optimization component, records with inadequate methodological detail (short notes and extended abstracts), non-English items, theses/patents/white papers without archival status, and duplicate or superseded versions. Distributed learning and parallel learning did not serve as a principal term, as it largely retrieves generic distributed training, which adds a lot of noise. The use of those terms in relevant works, but actually applying federated or privacy-preserving techniques, was eventually retrieved through backward and forward citation chasing. These criteria yielded the final corpus of 124 studies used in the PRISMA synthesis.

3.3 | Quality Assessment

Since the corpus consists, in the majority of cases, of algorithmic systems and simulation studies, instead of clinical intervention trials, formal clinical risk-of-bias tools were not used. Rather, a qualitative assessment of each study was performed on the basis of methodological clarity, datasets and preprocessing description, adequacy of evaluation metrics, and code or implementation information. The articles with significant reporting gaps were also part of the narrative synthesis but were not prioritized in the comparative tables and highlighted as the exemplar frameworks.

In order to enable reproducible and comparable findings, healthcare FL studies need to provide not only the predictive performance but also the significant methodological and system information. This should at least consist of datasets and preprocessing (train–validation–test splits), federated setup (number and type of clients, cross-silo or cross-device topology, and communication rounds), and complete hyperparameter configuration. The authors should publish code or configuration files where they can, along with fundamental runtime information (hardware and software versions). Privacy budget reporting (e.g., use of DP), communication cost, and, where possible, energy or latency measurements will also make fair benchmarking and meta-analysis in healthcare FL a possibility.

4 | Thematic Taxonomy and Literature Analysis

4.1 | Privacy and Security Approaches

In this section, the review summarizes the privacy and security state of FL in healthcare in accordance with the thematic taxonomy as a whole. The subsections are arranged like a hierarchy of privacy protection of the decentralized model training, the bottom one being statistical privacy, and the top one being governance and auditability. The initial theme, the DP and SA, gives statistical guarantees, ensuring the safety of individual records and limiting the server-side inference, which is why this topic is the most frequently used privacy mechanism in functioning FL systems. The second theme HE and multikey HE extend the protection into the computational space where an encrypted computation can be carried out across institutional data silos without decryption, thus maintaining confidentiality in the course of joint training. Third, there is the theme of TEEs, which addresses integrity at runtime by executing aggregation and decryption in hardware-isolated enclaves, which provide

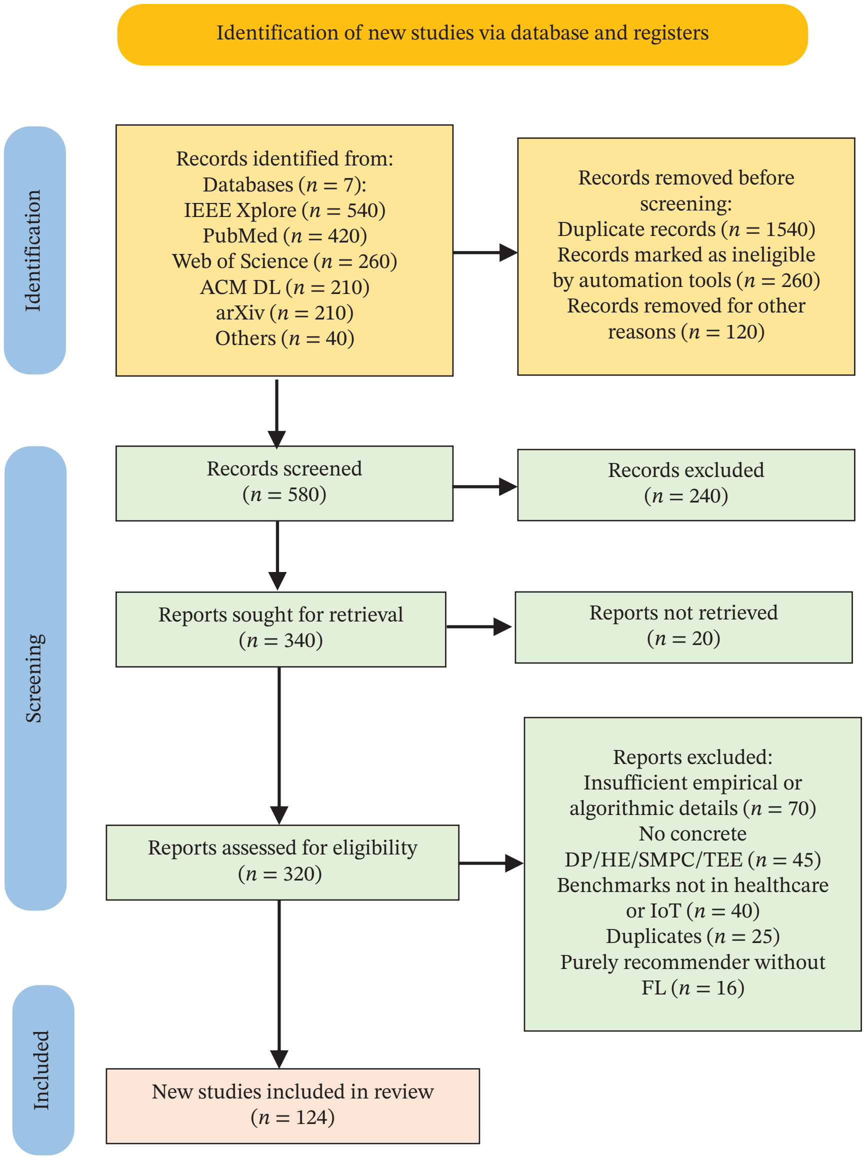

FIGURE 9 | PRISMA flow diagram.

verifiable guarantees of integrity of both code and data. The next group of themes is blockchain-based integrity, PETs, and systems of governance supporting human-compliant, transparent, and auditable healthcare FL deployments.

Although blockchain has the ability to enhance the integrity, auditability, and traceability in FL, its use in healthcare is not without constraints. On-chain logging may be an extra latency and storage cost that is impractical in a time-sensitive or limited-resource environment, as well as consensus protocols. In addition, regulatory and governance issues, including the interactions between the on-chain model metadata and the health-data protection legislations and institutional audit, are unresolved. In turn, blockchain-enhanced FL can be regarded as an additional layer, the benefits of which should be effectively balanced with the performance, complexity, and regulatory limitations of a particular healthcare implementation. Figure 10 summarizes the layered privacy and optimization stack in healthcare FL, linking cryptographic PETs, personalization mechanisms, and system-level orchestration. Table 2 maps the privacy and security approaches with their outcomes.

4.2 | Personalization and Optimization in FL

Personalization and optimization in FL are critical prerequisites for practical deployment in real-world healthcare settings. Clients in healthcare (wearables, mobile apps, and hospitals) present non-IID, heterogeneous data distributions, varying connectivity, computation, and trust levels. Standard aggregation methods such as FedAvg often degrade under such heterogeneity. Consequently, algorithmic innovations aim to (i) mitigate divergence (e.g., FedProx, FedNova, and SCAFFOLD), (ii) enable client-specific fine-tuning or adapter layers (personalization), (iii) adapt to system constraints (client capacity, dropouts, stragglers), and (iv) reduce communication and computation overhead while maintaining high accuracy.

In healthcare recommender and predictive systems, notions such as meta-learning for personalization, local adapter modules, layerwise fine-tuning, and client-specific model heads are increasingly reported. Optimization frameworks now include adaptive aggregation weights, learning rate scheduling per client, clipping and normalization for variance reduction, and multi-objective trade-off controllers that balance accuracy, fairness, latency, and communication cost. This section provides a structured literature analysis of these developments. Figure 11 provides a comparative schematic of different personalized and optimized FL algorithms.

The contents of Section 4.2 are organized to provide a coherent taxonomy of optimization and personalization strategies in FL, reflecting the evolution from foundational convergence algorithms to adaptive, decentralized, and fine-tuned personalization approaches. The FedAvg, FedProx, FedNova, SCAFFOLD, and FedOpt cluster represents the Fed backbone of FL that balances training in heterogeneous clients of healthcare by addressing client drift and global goals consistency. The second layer, FedPer, FedMeta-Health, FedAMP, FedBN, and FedDyn cluster, is the personalization layer, which allows meta-learning, layer-specific fine-tuning, and attention-based aggregation of clinically meaningful customization to patients, hospitals, or devices. The third cluster incorporates system-adaptive and resource-efficient models that control the cost of communication, client dropout, and energy limits, which play a critical role in lifelong learning in wearables and IoMT settings. Fourth cluster puts emphasis on the hierarchical and multitask designs in which edge-cloud

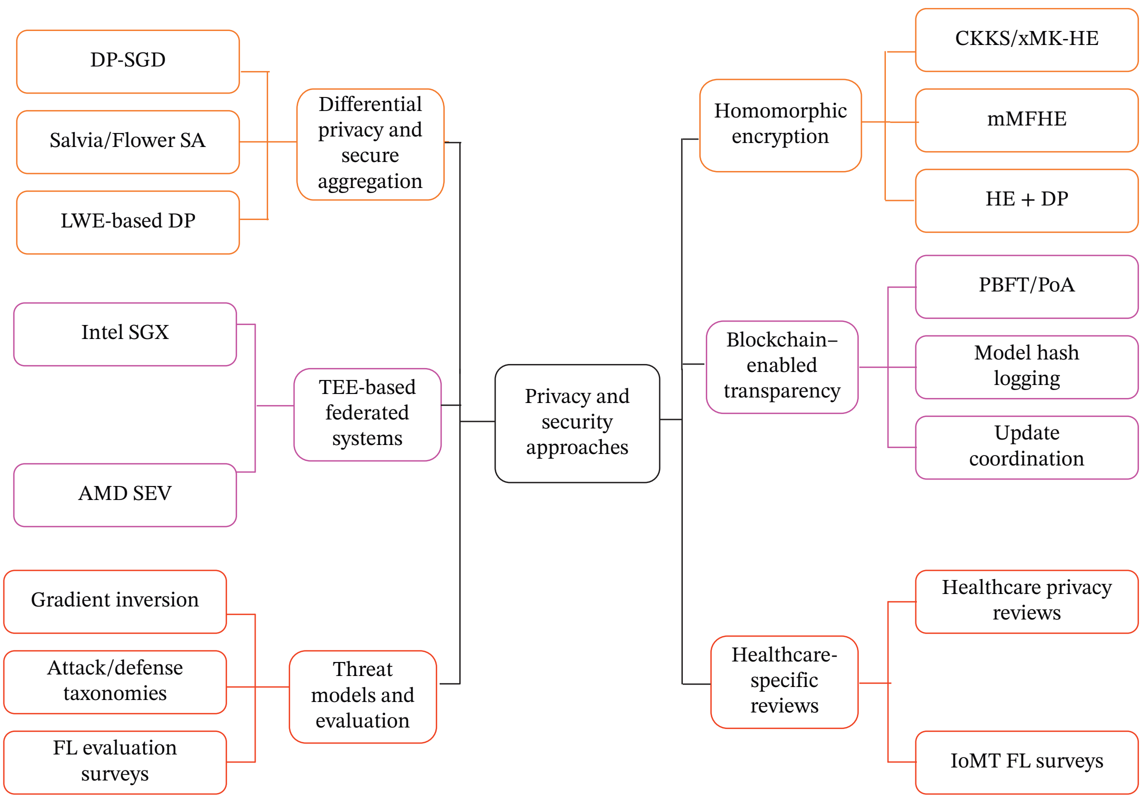

FIGURE 10 | Taxonomy of privacy and security approaches.

TABLE 2 | Mapping of privacy and security approaches with outcomes.

Cluster	Technique/mechanism	Domain/dataset	Representative outcome	References
Differential privacy and secure aggregation frameworks	DP-SGD with clipping + Salvia/Flower SA protocols; LWE-based DP aggregation	General FL frameworks; healthcare benchmark datasets	Scalable privacy; $\leq$ 3% accuracy loss vs. non-DP; robust dropout handling	[31, 32, 41, 43], [44]
Homomorphic encryption and multikey HE for healthcare FL	CKKS/xMK-HE, mMFHE, HE + DP	EHR and medical image tasks via Flower & HE libraries	Encrypted training with 2×–3× compute cost but no data exposure	[33–36, 42]
TEE-based federated systems	Intel SGX/AMD SEV enclaves for secure aggregation and model execution	Cross-silo hospital FL; edge healthcare nodes	End-to-end runtime integrity; near-native latency overhead	[38, 40]
Blockchain-enabled transparency and auditability	PBFT/PoA chains for model hash logging and update coordination (+HE)	IoT and clinical data sharing across institutions	Immutable provenance records; trusted coordination w/o central server	[45–48]
Threat models and evaluation protocols	Gradient inversion SoK, attack/defense taxonomies, FL evaluation surveys	Generic FL security benchmarks incl. healthcare	Clarifies attack surface ⟶ motivates defense-in-depth architecture	[49–51]
Healthcare-specific privacy reviews and IoMT FL surveys	Systematic reviews of FL applications in healthcare privacy and IoMT	MIMIC-IV, NIH chest X-ray, IoMT datasets	Consolidated design guidelines for secure health FL	[41, 52]

coordination and task-specific optimization enhance the fairness and scalability within the institutions. The fifth cluster is a graph and peer-to-peer topology, which decentralize the collaboration beyond the conventional client−server framework to promote resilient and privacy-preserving interactions. Lastly, the sixth cluster on self-supervised and fine-tuning schemes represents the recent trend of the hybrid global−local adaptation that optimizes the generalization and the patient sensitivity. Table 3 shows the mapping of personalization and optimization approaches with outcomes.

4.3 | Multimodal Data Fusion

This section is separated into six groups, which summarize the evolution of multimodal FL in medical practice. Theoretical foundations, foundational surveys, and taxonomies define early, intermediate, and late fusion strategies. Handling of incomplete or incongruent modalities addresses the problem of missing or mismatched modalities by using adaptive weighting and modality-invariant representations. Secure fusion and privacy-preserving methods use a combination of cryptographic and DP systems to secure sensitive multimodal healthcare information. Cross-modal learning and representation alignment are semantic integration methods that systematically contrast and attentively train imaging, text, and EHR modalities. Communication-efficient and system-level MFL provides the meaning of scalability and real-time applicability by minimizing communication and energy overhead in IoMT environments. Lastly, the practical implementation of domain-specific multimodal applications in medical report generation and tumor segmentation shows that the clinical utility of multimodal fusion is justified, given federated conditions. Figure 12 provides insights into multimodal data fusion techniques. Table 4 provides the techniques, datasets, mechanisms, and outcomes of each theme. Table 4 shows the mapping of multimodal data fusion approaches with outcomes.

4.4 | Communication Efficiency and Scalability

Large-scale healthcare FL must operate under limited bandwidth, heterogeneous devices, client dropouts, and variable latency. Communication becomes the dominant bottleneck as models and cohorts scale across hospitals, wearables, and IoMT nodes. As shown in Figure 13, the literature converges on six complementary levers: (i) hierarchical (edge–cloud) aggregation, (ii) asynchronous/semiasynchronous updates to hide stragglers, (iii) client selection and resource allocation, (iv) gradient/model compression (quantization, sparsification, and low-rank), (v) hybrid H/V-FL partitioning for data alignment, and (vi) end-to-end systems that co-optimize accuracy, latency, and energy. Recent surveys and systems studies (2022–2025) show that combining hierarchical topologies with compression and informed client selection yields the most robust gains in rounds-to-convergence and total bytes transmitted while maintaining accuracy in health tasks.

Hierarchical and edge-cloud FL emphasizes topological scaling through multitier aggregation. Asynchronous and client selection strategies capture time- and participation-adaptive scheduling for large, heterogeneous networks. Gradient compression and quantization unite methods that directly reduce transmitted bytes. Hybrid and cross-topology FL highlights federations combining horizontal and vertical data partitions. Energy-

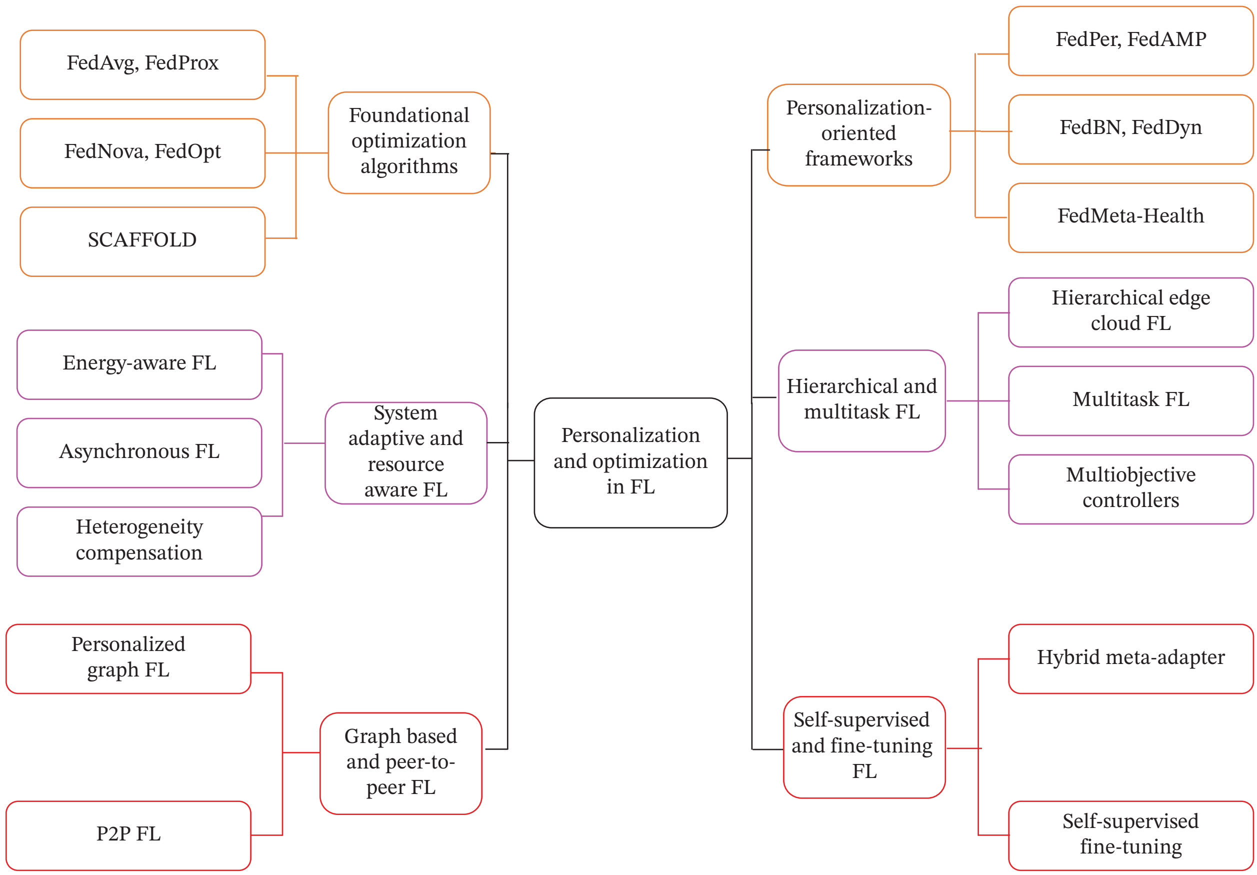

FIGURE 11 | Taxonomy of personalization and optimization approaches.

efficient and scalable IoT systems address real-time and low-power deployments, while communication-efficient frameworks and surveys synthesize overarching taxonomies. Table 5 shows the mapping of communication efficiency and scalability approaches.

5 | Comparative Study

The comparative analysis presented in this section is beyond reporting accuracy to bring out the trade-offs in privacy, personalization, multimodality, and communication. Methods that are efficient in communication such as FedCSTQ and hybrid H/V-FL are also more scalable and save on bandwidth, but possibly at the cost of accuracy or personalization in highly heterogeneous environments, with multimodal frameworks such as multimodal FL fusion increasing accuracy and personalization but at the cost of more architectural complexity. These differences illustrate the fact that there is no one model that prevails in all dimensions, and the decision about the method should be adjusted in accordance with the clinical task, infrastructure, and regulatory needs.

The comparative study consolidates the findings from the previous thematic analyses (Sections 4.1–4.4) to provide an integrated view of how FL frameworks have evolved in healthcare. Although many algorithms exist, evaluation is conducted across different datasets, privacy measures, and communication environments. Table 6 levels the comparison with evaluation parameters such as accuracy, privacy protection, communication efficiency, personalization, and scalability to point out trade-offs between representative FL frameworks. The aim is to find structures that strike a balance between technical efficiency and healthcare applicability and uncover the research gap to design the next generation.

Figure 14 compares the accuracy of the classification of nine FL frameworks in the bar chart. The highest accuracy is 93.2% of multimodal FL fusion, then FedBN + FedCSTQ (92.0%), CHPFL (91.6%), and then a relatively low but still competitive performance of FedProx + DP, SCAFFOLD, and FedCSTQ in the 86%–88% range.

Figure 15 represents a radar graph that compares FL frameworks using modality support, privacy mechanisms, and scalability. Visualization is made with normalization of a score; modality (unimodal = 0.5 and multimodal = 1.0), privacy strength (DP = 0.25, SA = 0.5, intermediate mechanisms = 0.75, and hybrid mechanisms = 1.0) and scalability (medium = 0.5, high = 0.75, and very high = 1.0). It is easier to make comparisons and interpretations across frameworks based on these values.

6 | Identified Gaps and Challenges

This section, which is based on the comparison above, takes a more critical approach and dwells upon the areas in which the current healthcare FL approaches fall short. Most of the algorithms which do well in controlled benchmarks continue to fail when using real non-IID client distributions, changing data regimes, stringent privacy constraints, and limited resources. The subsequent subsections thus focus on these gaps that remain

TABLE 3 | Mapping of personalization and optimization approaches with outcomes.

Cluster	Representative techniques	Domain/dataset	Mechanism	Key outcomes	References
Foundational optimization algorithms	FedAvg + FedProx + FedNova + SCAFFOLD + FedOpt	Generic FL frameworks; medical and IoT data	Gradient normalization, proximal loss, adaptive server optimizers	Improved stability and convergence under non-IID data	[53–56]
Personalization-oriented frameworks	FedPer, FedMeta-Health, FedAMP, FedBN, FedDyn	EHR, medical imaging, mHealth recommender	Meta-learning, local batch-norm, adaptive regularization, attention weights	Personalized models with $\leq$ 3% accuracy loss vs. centralized training	[37, 57–61]
System-adaptive and resource-aware FL	Energy-aware FL, asynchronous FL, heterogeneity compensation	Wearable and IoT-health networks	Dynamic client selection, adaptive LR, dropout mitigation	Lower energy, communication and latency costs; maintained model accuracy	[62–65]
Hierarchical and multitask FL	Hierarchical edge-cloud FL multitask FL multiobjective controllers	Hospital–edge ecosystems, multidisease datasets	Multilevel aggregation, task-specific optimization, Pareto control	Faster convergence and improved fairness across tasks	[66–68]
Graph-based and peer-to-peer FL	Personalized graph FL P2P FL	Health IoT, social-health graphs	Graph aggregation and peer model exchange	Decentralized personalization with robustness to server faults	[69, 70]
Self-supervised and fine-tuning FL	Hybrid meta-adapter self-supervised fine-tuning	Wearable and passive sensing datasets	Global pretraining + local fine-tuning	Rapid adaptation with minimal local computation	[71, 72]

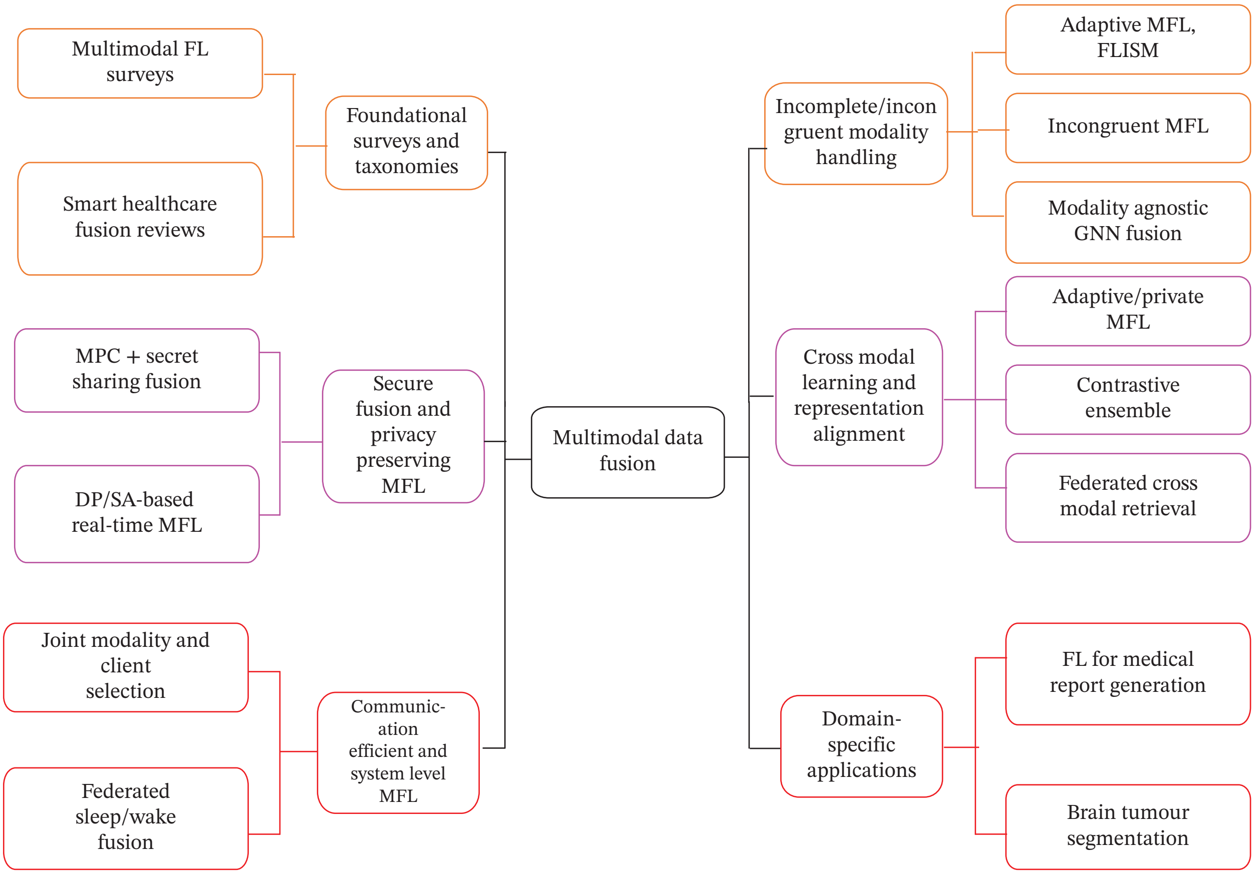

FIGURE 12 | Taxonomy of multimodal data fusion approaches.

unanswered instead of repeating what has been already known to be advantageous. FL in medicine continues to be limited in terms of research and operational issues that restrict the application to a clinical scale. The above comparative reviews have shown that most of the frameworks work well within controlled benchmarks but fail to work well in medical scenarios where data distributions, operating capabilities of devices, and policies of an institute are dramatically different. This section points out the significant gaps and technological limitations that remain to be the sources of reliability, interpretability, and fair performance in the case of healthcare-oriented FL.

6.1 | Data Heterogeneity

One of the biggest challenges of federated healthcare systems is data heterogeneity. The data produced by hospitals, clinics, and IoMT devices have different formats, resolutions, and statistical characteristics. The non-IID distribution of clients might worsen global convergence and cause bias [53, 59]. Algorithms such as FedProx, FedBN, and CHPFL mitigate this drift by regularizing proximal or localized normalization but cannot cope with domain shifts when they become more severe or imaging modalities become even more different [96, 103]. Even multimodal fusion systems such as MFL fusion [86] use the alignment of cross-modal features; the absence of other modalities or uncontrolled site sampling can cause model updates to be unstable and contribute to a greater gradient variance.

6.2 | Privacy-Utility Trade-Offs

There is an additional point of conflict in the privacy-utility trade-off. DP and SA guarantee high levels of confidentiality, but tend to add noise or cryptographic overhead to models, which reduces model accuracy [109]. Recent compression-based schemes help to mitigate this cost [112], but the trade-offs between the encryption strength and the computational efficiency are yet to be considered in a real-time context involving healthcare. In hybrid solutions involving DP with trusted execution or secret sharing, partial improvements are obtained [96], however, at the scale of pilot deployments only [113].

6.3 | Personalization Drift

An even more subtle and yet equally problematic concern is personalization drift, in which local adaptation in models tailored to patients or institutions causes uneven worldwide learning. Models such as FedAMP and FedBN are more personalized by isolating or weighting local layers [59, 60] only to be reintroduced by retraining and frequent global aggregation. This issue becomes severe in long-term hospital chain networks where data streams change, meaning that adaptive federated meta-learning or drift-aware controllers are necessary [114]. Altogether, the interaction of heterogeneous data, privacy enforcement, and stability of personalization forms the future horizon of federated healthcare research.

TABLE 4 | Mapping of multimodal data fusion approaches with outcomes.

Cluster	Representative techniques	Domain/dataset	Mechanism	Outcome	References
Foundational surveys and taxonomies	Multimodal FL surveys, smart healthcare fusion reviews	Healthcare and IoT	Early/late/intermediate fusion taxonomy; modality heterogeneity analysis	Established conceptual foundation and research gaps	[73–77]
Incomplete/incongruent modality Handling	Adaptive MFL, FLISM, incongruent MFL, modality-agnostic GNN fusion	Wearables, sensor and imaging data	Modality-invariant encoding, quality-aware aggregation, and imputation of missing modalities	Improved robustness (+6–10% accuracy under missing modalities)	[78–83]
Secure fusion and privacy-preserving MFL	MPC + secret-sharing fusion, DP/SA-based real-time MFL	EHR + sensor streams	Cryptographic fusion with DP noise and secure sharing	Achieved privacy guarantees with negligible utility loss	[84, 85]
Cross-modal learning and representation alignment	Adaptive/private MFL, contrastive ensemble, federated cross-modal retrieval	Imaging + text + EHR	Cross-modal attention and contrastive embedding alignment	Enhanced retrieval/report generation; 5%–8% AUC gain vs. unimodal FL	[86–88]
Communication-efficient and system-level MFL	Joint modality and client selection, federated sleep/wake fusion	IoMT and wearables	Selective modality aggregation, asynchronous updates	Reduced bandwidth (−30%) w/similar accuracy	[89, 90]
Domain-specific applications	FL for medical report generation, brain tumor segmentation	Clinical imaging + text + genomic data	Task-specific multimodal fusion under FL	Demonstrated high task-specific accuracy	[91, 92]

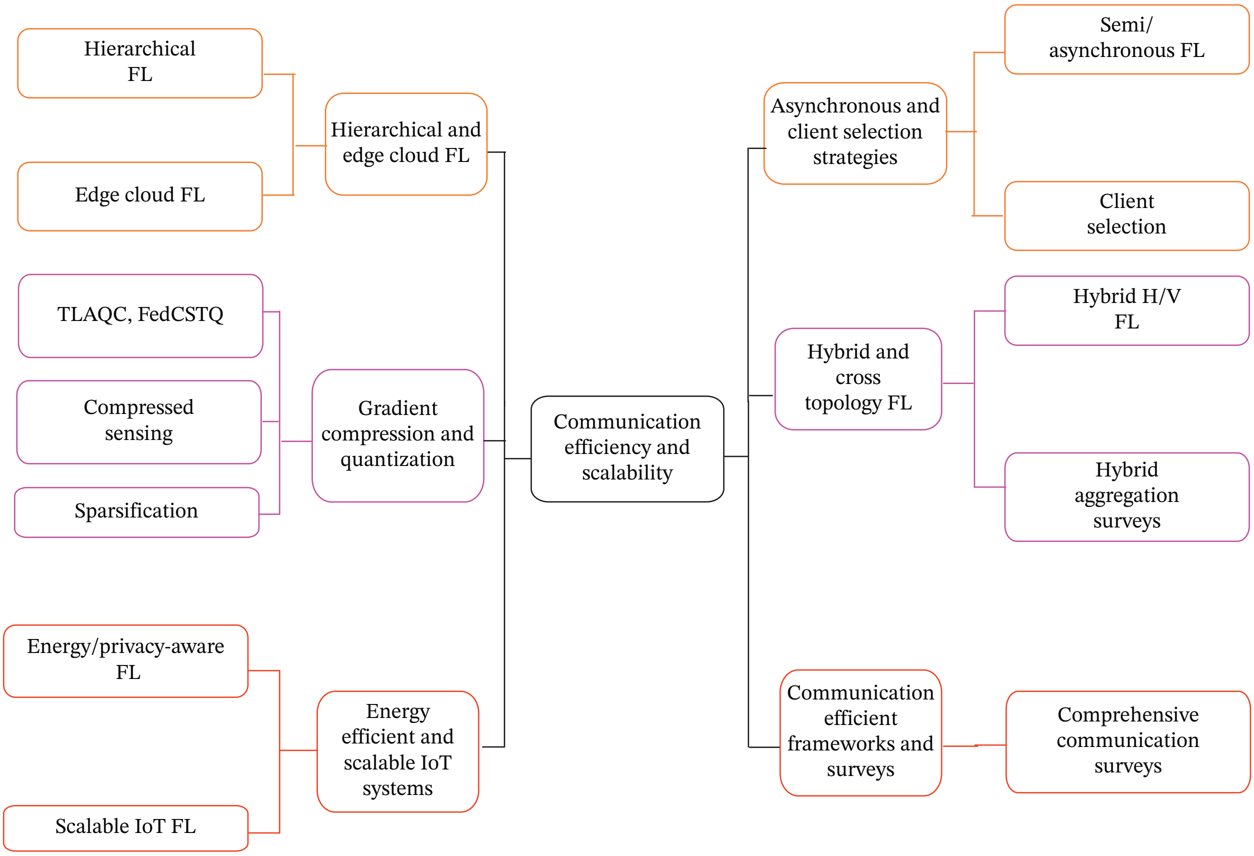

FIGURE 13 | Taxonomy of communication efficiency and scalability approaches.

6.4 | Real-World Healthcare Deployments and Initiatives

FL has already been piloted on a number of real-world healthcare environments that have demonstrated the potential as well as the challenges it faces in practice. FL can enable multi-institutional medical imaging and clinical outcome prediction for multi-institutions to attain competitive performance without centralizing patient information [1–3, 5]. As an illustration, MRI and CT consortia training have been cross-trained on brain tumor and COVID-19 imaging, maintaining the data sovereignty of local data [1, 2], and federated models have been employed to forecast the outcome of COVID-19 across dispersed hospitals [3]. Simultaneously, the edge device–based federated physiological monitoring and anomaly detection in the smaller-scale deployments in smart healthcare and IoMT settings have been examined in smaller-scale deployments. Such efforts note that technical progress in the privacy, personalization, and communication domains should be supported by strong governance, operational processes, and clinical validation guidelines to make FL progressively integrated into hospital information systems and digital health services.

7 | Future Research Directions

In the future of healthcare FL, convergence of edge intelligence, quantum-resilient security, and sustainable deployment frameworks will evolve the next generation of healthcare FL. Edge intelligence brings the computation nearer to data sources, including wearable sensors, medical imaging machines, and hospital IoT gateways. The combination of edge computing and FL mitigates communication latency, enhances privacy, and supports real-time decision-making in telehealth and intensive care monitoring. Several current research studies have shown the potential of Edge-FL in achieving reduced latency and enhanced adaptability among hospital networks and mobile devices [115, 116]. By hierarchically coordinating edge and cloud, hierarchical orchestration can also help optimize workloads by partitioning model layers to be trained locally and synchronized globally [117].

The future of security in FL is quantum-safe FL. Classical cryptographic tools such as DP and HE will be exposed to quantum adversaries. Model aggregation in FL can be secured against quantum attacks by integrating postquantum cryptography (PQC) and quantum-key distribution (QKD) in the system [118]. Quantum FL (QFL), early models of quantum circuits to train models in parallel and enjoy better privacy, demonstrates both the benefits of speed and resiliency [119, 120]. The difficulty here is that hybrid quantum–classical FL frameworks need to be developed, which can be interoperable with current medical infrastructure and data protection regulations, including the HIPAA and GDPR. Healthcare-related recommendation is also a significant advantage to patients. The addition of an optimized recommender system into medical applications and a privacy-preserving model will be a game-changer.

Along with the development of new architectures and privacy mechanisms, work in the future must take a more systematic evaluation protocol. In predictive work, the outcomes of the study are reported not only with the accuracy and AUC but with

TABLE 5 | Mapping of communication efficiency and scalability approaches with outcomes.

Cluster	Representative techniques	Domain/dataset	Mechanism	Key outcomes	References
Hierarchical and edge–cloud FL	Hierarchical FL	IoT-health/ edge-cloud	Multitier aggregation; edge personalization	25%–35% reduction in rounds and latency	[93–97]
Asynchronous and client selection strategies	Semi-/asynchronous FL, client selection	IoT, mobile health	Bounded staleness + reward-based participation	Improved throughput and fairness; lower straggler delay	[98–106]
Gradient compression and quantization	TLAQC, FedCSTQ, compressed sensing, sparsification	IoT/wireless FL	Gradient sparsification, ternary quantization	$\geq$ 40% communication reduction, stable accuracy	[102, 107–109]
Hybrid and cross-topology FL	Hybrid H/V-FL, hybrid aggregation surveys	E-Health systems	Horizontal + vertical fusion with adaptive update shrinking	Maintained accuracy; up to 30% bandwidth saving	[103, 104]
Energy-efficient and scalable IoT systems	Energy/privacy-aware FL, scalable IoT FL	Wearable/sensor networks	Co-optimization of energy, latency, communication	Up to 45% power savings with similar accuracy	[93, 110, 111], [104, 112]
Communication-efficient frameworks and surveys	Comprehensive comm. surveys	Cross-domain	Taxonomies, frameworks, and benchmark analyses	Defined open challenges, best practices	[98, 100, 112]

TABLE 6 | Comparison of existing frameworks and datasets.

Framework	Year	Dataset/domain	Modality	Privacy mechanism	Optimization/communication	Accuracy (%)	Scalability	References
FedProx + DP	2021	MIMIC-III (EHR)	Unimodal	DP	Proximal term to stabilize client drift	87.2	Medium	[53]
SCAFFOLD	2021	IoT-Health simulation	Unimodal	SA	Variance reduction via control variates	86.5	Medium	[54]
FedBN	2024	MRI + CT (imaging)	Multimodal	Local BatchNorm isolation	Hierarchical aggregation	90.8	High	[59]
FedAMP	2024	EHR + IoT sensors	Multimodal	Attention-based personalization	Adaptive client weighting	88.9	High	[60]
FedCSTQ	2024	IoT Healthcare	Unimodal	HE + ternary quantization	Compressed sensing + sparse updates	86.5	Very High	[109]
CHPFL	2025	Edge-cloud (clinical IoT)	Multimodal	Hybrid DP + SMPC	Clustered hierarchical aggregation	91.6	High	[96]
Hybrid H/V-FL	2024	E-Health (partitioned data)	Multimodal	SA	Adaptive shrinking + hybrid topology	89.7	High	[103]
FedBN + FedCSTQ (combined)	2025	Smart hospital edge	Multimodal	HE + local BatchNorm	Compression + hierarchical synchronization	92.0	Very high	[59, 109]
Multimodal FL fusion	2025	EHR + imaging + text	Multimodal	DP + cross-modal attention	Adaptive fusion with client alignment	93.2	High	[86]

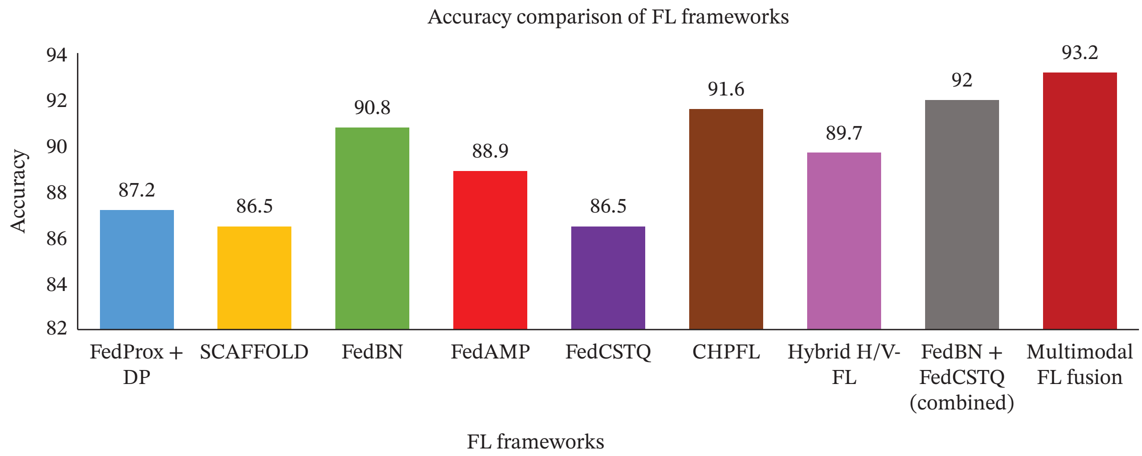

FIGURE 14 | Comparative performance of FL frameworks across key metrics.

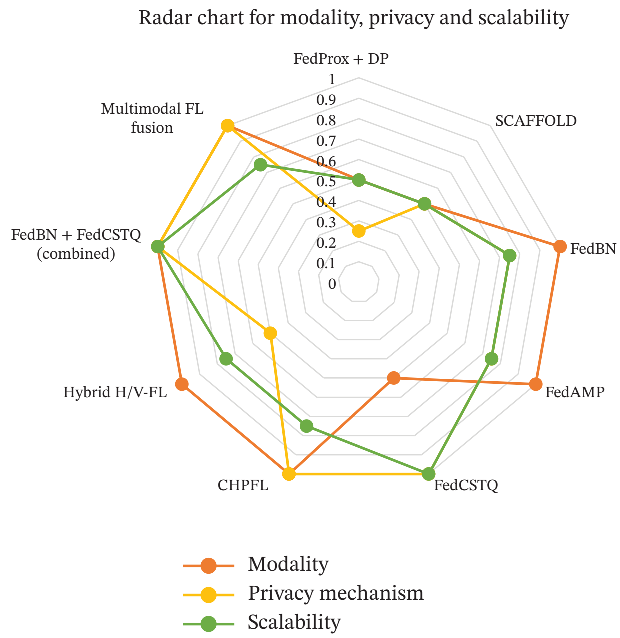

FIGURE 15 | Radar chart of FL frameworks.

the task-specific clinical performance (sensitivity, specificity, precision−recall and, when feasible, the quality of recall at k or top-k recommendations). In the case of privacy-preserving methods, the privacy budget (e) along with any composition or accounting assumptions is to be specified. Communication level contribution reported communication cost (e.g., bytes per client per round and number of rounds) and end-to-end latency, and, where available, energy consumption on representative devices is encouraged. With the help of this multidimensional system of metrics, it will be possible to compare FL methods more clearly and correspond to clinical and operational needs in a healthcare environment with more accuracy.

Lastly, sustainability is becoming a matter of concern as a critical aspect of healthcare FL. The energy and carbon footprint of large-scale FL necessitates the co-optimization of communication, energy consumption, and hardware efficiency [121]. Energy-conscious selection of clients, model compression, and recyclable-powered edge clusters are among the strategies that allow deploying FL in an environmentally friendly way [122]. In addition, a lifecycle governance approach is necessary to maintain the legibility, interpretability, and recyclability of federated models in clinical ecosystems [123]. In addition, the combination of edge intelligence to provide responsiveness, quantum-safe encryption to ensure long-term reliability, and green computing principles to ensure environmental and operational sustainability will lead to scalable, secure, and sustainable FL in healthcare [124]. Future studies ought to build adaptive standards that could analyze energy, privacy, and accuracy collectively to enhance responsible and strong healthcare AI ecosystems.

8 | Conclusion

The review offers a synthesis of the latest trends in privacy-preserving FL in healthcare, tracing the research area from 2020 to 2025 across various technical and application aspects. Important thematic areas, such as privacy and security mechanisms, personalization and optimization, multimodal data fusion, and communication efficiency, were incorporated into the discussion based on over 124 authentic publications. Relative analysis showed that frameworks that include FedBN, FedCSTQ, CHPFL, and MFL fusion are advancing, balancing model accuracy, privacy protection, and communication scalability. Nevertheless, even with such successes, there are still current challenges. In the real world, deployment is still limited due to issues of data heterogeneity, compromise of privacy-utility, drift in personalization, and limited interpretability, particularly when deployed in multi-institutional hospital settings and to nonhomogenous IoMT ecosystems.

The results underline the fact that the following generation of healthcare FL should not be limited to the isolated algorithmic innovation, but the system-level intelligence, involving edge computing, quantum-safe cryptography, and sustainable model lifecycle control. The incorporation of edge intelligence will minimize the latency and improve the responsiveness of time-constrained clinical processes, and the mechanisms of post-quantum encryption and hybrid encryption will guarantee the long-term safety of information against future threats posed by computational attacks. In terms of implementation, the adoption of communication-efficient, energy-conscious protocols will also make federated systems in line with global sustainability goals. These findings will offer a single comparative framework to researchers and practitioners to assess the interoperability and ethical preparedness of FL frameworks in regulated healthcare structures.

To achieve scientific reproducibility and credibility, in future research, standardized datasets, publicly available code repositories and transparent descriptions of runtime environments, dynamic hyperparameters, and evaluation metrics should be used. Self-reporting on energy use, privacy budgets, and communication rounds, as well as performance indicators, will further enable fair benchmarking. Eventually, reproducible, safe, and sustainable FL will be the basis for reliable digital healthcare, supporting collaborative intelligence without undermining privacy, efficiency, or ethical responsibility.

Nomenclature

AI	Artificial intelligence
ML	Machine learning
DL	Deep learning
FL	Federated learning
HFL	Hierarchical federated learning
H/V-FL	Horizontal/vertical federated learning
PFL	Personalized federated learning
IoT	Internet of things
IoMT	Internet of medical things
EHR	Electronic health record
EMR	Electronic medical record
ICU	Intensive care unit
PETs	Privacy-enhancing technologies
DP	Differential privacy
HE	Homomorphic encryption
FHE	Fully homomorphic encryption
SA	Secure aggregation
MPC/ SMPC	(Secure) multiparty computation
TEE	Trusted execution environment
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
AUC	Area under the curve
ROC	Receiver operating characteristic
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RQ	Research question
FedAvg	Federated averaging
FedProx	Federated proximal
FedNova	Normalized averaging federated optimization
FedBN	Federated batch normalization
FedAMP	Federated attentive message passing
FedCSTQ	Federated learning with compressed sensing and ternary quantization
CHPFL	Clustered hierarchical personalized federated learning
MIMIC	Medical information mart for intensive care
TLAQC	Two-layer accumulated quantized compression
FLISM	Federated learning with incomplete sensing modalities
FedDyn	Federated dynamic
SCAFFOLD	Stochastic controlled averaging for federated learning
FedOpt	Federated optimization (FedAdam/FedYogi/ FedAdagrad family)
FedPer	Personalized federated learning
DP-SGD	Differentially private stochastic gradient descent
Intel SGX	Intel software guard extension
AMD SEV	AMD secure encrypted virtualization
CKKS	Cheon–Kim–Kim–Song scheme
xMK-HE	Extended multikey homomorphic encryption
mMFHE	Multiparty/multikey fully homomorphic encryption
PBFT	Practical Byzantine fault tolerance
PoA	Proof-of-authority

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

References

1. N. Rieke, J. Hancox, W. Li, et al., "The Future of Digital Health with Federated Learning," *NPJ Digital Medicine* 3, no. 1 (2020): 119, https://doi.org/10.1038/s41746-020-00323-1.

2. M. J. Sheller, B. Edwards, G. A. Reina, et al., "Federated Learning in Medicine: Facilitating multi-institutional Collaborations Without Sharing Patient Data," *Scientific Reports* 10, no. 1 (2020): https://doi.org/10.1038/s41598-020-69250-1.

3. I. Dayan, H. R. Roth, A. Zhong, et al., "Federated Learning for Predicting Clinical Outcomes in Patients with COVID-19," *Nature Medicine* 27, no. 10 (2021): 1735–1743, https://doi.org/10.1038/s41591-021-01506-3.

4. J. Wicaksana, Z. Yan, X. Yang, Y. Liu, L. Fan, and K.-T. Cheng, "Customized Federated Learning for Multi-Source Decentralized Medical Image Classification," *IEEE Journal of Biomedical and Health Informatics* 26, no. 11 (2022): 5596–5607, https://doi.org/10.1109/JBHI.2022.3198440.

5. Z. L. Teo, L. Jin, S. Li, et al., "Federated Machine Learning in Healthcare: a Systematic Review on Clinical Applications and Technical Architecture," *Cell Reports Medicine* 5, no. 2 (2024): https://doi.org/10.1016/j.xcrm.2024.101419.

6. A. E. W. Johnson, T. J. Pollard, L. Shen, et al., *MIMIC-IV* (PhysioNet, 2024), https://doi.org/10.13026/s6n6-xd98.

7. P. Kairouz, H. B. McMahan, B. Avent, et al., "Advances and Open Problems in Federated Learning," *Found Trends Mach Learn* 14, no. 1–2 (2021): 1–210, https://doi.org/10.1561/2200000083.

8. S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic Controlled Averaging for Federated Learning," in *Proceedings of the 37th International Conference on Machine Learning. Vol 119. Proceedings of Machine Learning Research*, ed. H. D. Iii and A. Singh (PMLR, July 2020), 5132–5143, https://doi.org/10.48550/arXiv.1910.06378.

9. J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, "Tackling the Objective Inconsistency Problem in Heterogeneous Federated Optimization," *Advances in Neural Information Processing Systems* 33 (2020): 7611–7623, https://doi.org/10.48550/arXiv.2007.07481.

10. T. Jin, S. Pan, X. Li, and S. Chen, "Metadata and Image Features Co-Aware Personalized Federated Learning for Smart Healthcare," *IEEE Journal of Biomedical and Health Informatics* 27, no. 8 (2023): 4110–4119, https://doi.org/10.1109/JBHI.2023.3279096.

11. K. H. Li, P. P. B. de Gusmão, D. J. Beutel, and N. D. Lane, "Secure Aggregation for Federated Learning in Flower," in *Proceedings of the 2nd ACM International Workshop on Distributed Machine Learning* (December 2021), 8–14, https://doi.org/10.48550/arXiv.2205.06117.

12. A. Yousefpour, I. Shilov, A. Sablayrolles, et al., "Opacus: User-Friendly Differential Privacy Library in PyTorch," *arXiv [csLG]* (2022).

13. D. J. Beutel, T. Topal, A. Mathur, et al., "Flower: a Friendly Federated Learning Research Framework," *arXiv [csLG]* (2022).

14. N. Hernandez-Cruz, P. Saha, M. M. K. Sarker, and J. A. Noble, "Review of Federated Learning and Machine Learning-Based Methods for Medical Image Analysis," *Big Data and Cognitive Computing* 8, no. 9 (2024): 99, https://doi.org/10.3390/bdcc8090099.

15. K. Dasaradharami Reddy and T. R. Gadekallu, "A Comprehensive Survey on Federated Learning Techniques for Healthcare Informatics," *Computational Intelligence and Neuroscience* 2023, no. 1 (2023): https://doi.org/10.1155/2023/8393990.

16. S. Zhan, L. Huang, G. Luo, S. Zheng, Z. Gao, and H.-C. Chao, "A Review on Federated Learning Architectures for Privacy-Preserving AI: Lightweight and Secure cloud–edge–end Collaboration," *Electronics* 14, no. 13 (2025): 2512, https://doi.org/10.3390/electronics14132512.

17. T. Yanagi, S. Ikeda, N. Sukegawa, and Y. Takano, "Privacy-Preserving Recommender System Using the Data Collaboration Analysis for Distributed Datasets," *PLoS One* 20, no. 4 (2025): e0319954, https://doi.org/10.1371/journal.pone.0319954.

18. J. Wen, Z. Zhang, Y. Lan, Z. Cui, J. Cai, and W. Zhang, "A Survey on Federated Learning: Challenges and Applications," *International Journal of Machine Learning and Cybernetics* 14, no. 2 (2023): 513–535, https://doi.org/10.1007/s13042-022-01647-y.

19. J. Fu, Y. Hong, X. Ling, et al., "Differentially Private Federated Learning: a Systematic Review," *arXiv [csCR]* (2025).

20. R. Eden, I. Chukwudi, C. Bain, et al., "A Scoping Review of the Governance of Federated Learning in Healthcare," *NPJ Digital Media* 8, no. 1 (2025): 427, https://doi.org/10.1038/s41746-025-01836-3.

21. M. Li, P. Xu, J. Hu, Z. Tang, and G. Yang, "From Challenges and Pitfalls to Recommendations and Opportunities: Implementing Federated Learning in Healthcare," *arXiv [csLG]* (2025).

22. N. Rana and H. Marwaha, "Role of Federated Learning in Healthcare Systems: a Survey," *Mathematical Foundations of Computing* 7, no. 4 (2024): 459–484, https://doi.org/10.3934/mfc.2023023.

23. M. Asad, S. Shaukat, E. Javanmardi, J. Nakazato, and M. Tsukada, "A Comprehensive Survey on Privacy-Preserving Techniques in Federated Recommendation Systems," *Applied Sciences* 13, no. 10 (2023): 6201, https://doi.org/10.3390/app13106201.

24. M. Nasajpour, S. Pouriyeh, R. M. Parizi, et al., "Federated Learning in Smart Healthcare: a Survey of Applications, Challenges, and Future Directions," *Electronics* 14, no. 9 (2025): 1750, https://doi.org/10.3390/electronics14091750.

25. H. Zhang, F. Luo, J. Wu, X. He, and Y. Li, "LightFR: Lightweight Federated Recommendation with Privacy-preserving Matrix Factorization," *ACM Transactions on Information Systems* 41, no. 4 (2023): https://doi.org/10.1145/3578361.

26. H. Joshi and S. Joseph, "Standardization and Interoperability: Federated Learning's Impact on EHR Systems and Health Informatics," *Advances in Health Information Science and Practice* 1, no. 1 (2025): https://doi.org/10.63116/UBYM3803.

27. A. Sharma, N. K. Goswami, and B. Luitel, "Privacy Preserving Recommender Systems," *International Journal For Multidisciplinary Research* (2024): https://doi.org/10.36948/ijfmr.2024.v06i03.20301.

28. D. Chai, L. Wang, L. Yang, J. Zhang, K. Chen, and Q. Yang, "A Survey for Federated Learning Evaluations: Goals and Measures," *IEEE Transactions on Knowledge and Data Engineering* 36, no. 10 (2024): 5007–5024, https://doi.org/10.1109/TKDE.2024.3382002.

29. Z. Cai, T. Tang, S. Yu, Y. Xiao, and F. Xia, "Marking the Pace: a Blockchain-Enhanced Privacy-Traceable Strategy for Federated Recommender Systems," *arXiv [csLG]* (2024).

30. J. Luo, X. Yang, X. Yi, and F. Han, "Privacy-Preserving Recommendation System Based on User Classification," *Journal of Information Security and Applications* 79 (2023): https://doi.org/10.1016/j.jisa.2023.103630.

31. H. Fereidooni, S. Marchal, M. Miettinen, et al., "SAFELearn: Secure Aggregation for Private FEderated Learning," in *2021 IEEE Security and Privacy Workshops (SPW)* (San Francisco, CA, USA, August 2021), 56–62, https://doi.org/10.1109/SPW53761.2021.00017.

32. X. Zhang, Y. Luo, and T. Li, "A Review of Research on Secure Aggregation for Federated Learning," *Future Internet* 17, no. 7 (2025): 308, https://doi.org/10.3390/fi17070308.

33. I. Walskaar, M. C. Tran, and F. O. Catak, "A Practical Implementation of Medical Privacy-Preserving Federated Learning Using Multi-Key Homomorphic Encryption and Flower Framework," *Cryptography* 7, no. 4 (2023): 48, https://doi.org/10.3390/cryptography7040048.

34. H. R. Kumbhar and S. Srinivasa Rao, "Federated Learning Enabled multi-key Homomorphic Encryption," *Expert Systems With Applications* 268 (2025): https://doi.org/10.1016/j.eswa.2024.126197.

35. H. Kurniawan and M. Mambo, "Homomorphic Encryption-Based Federated Privacy Preservation for Deep Active Learning," *Entropy* 24, no. 11 (2022): 1545, https://doi.org/10.3390/e24111545.

36. J. Shen, Y. Zhao, S. Huang, and Y. Ren, "Secure and Flexible Privacy-Preserving Federated Learning Based on Multi-Key Fully Homomorphic Encryption," *Electronics* 13, no. 22 (2024): 4478, https://doi.org/10.3390/electronics13224478.

37. Y. Wang, H. Hu, and W. Du, "Communication-Efficient Personalized Federated Learning with Evolutionary Strategies and cluster-aware Knowledge Transfer," *Knowledge-Based Systems* 328 (2025): https://doi.org/10.1016/j.knosys.2025.114246.

38. L. Zhang, B. Duan, J. Li, Z. Ma, and X. Cao, "A TEE-Based Federated Privacy Protection Method: Proposal and Implementation," *Applied Sciences* 14, no. 8 (2024): 3533, https://doi.org/10.3390/app14083533.

39. M. Birgersson, C. Artho, and M. Balliu, "Sharing Without Showing: Secure Cloud Analytics with Trusted Execution Environments," *arXiv [csCR]* (2024).

40. T. Geppert, S. Deml, D. Sturzenegger, and N. Ebert, "Trusted Execution Environments: Applications and Organizational Challenges,"

Frontiers of Computer Science 4 (2022): https://doi.org/10.3389/fcomp.2022.930741.

41. S. T. Shah, Z. Ali, M. Waqar, and A. Kim, "Federated Learning in Public Health: a Systematic Review of Decentralized, Equitable, and Secure Disease Prevention Approaches," *Healthcare* 13, no. 21 (2025): 2760, https://doi.org/10.3390/healthcare13212760.

42. K. M. Babu, E. Bhavitha, M. Mythri, A. Anusha, B. Sri, and C. Gazala Akhtar, *Privacy-Preserving Federated Learning for Healthcare: A Synergistic Approach Using Differential Privacy and Homomorphic Encryption* (2024), https://doi.org/10.2139/ssrn.5088943.

43. T. Stevens, C. Skalka, C. Vincent, J. Ring, S. Clark, and J. Near, "Efficient Differentially Private Secure Aggregation for Federated Learning via Hardness of Learning with Errors," *arXiv [csCR]* (2021).

44. C. Yan, X. Lu, P. Lio, and P. Hui, "EARVP: Efficient Aggregation for Federated Learning with Robustness, Verifiability, and Privacy," *IEEE Transactions on Information Forensics and Security* 20 (2025): 5936–5951, https://doi.org/10.1109/TIFS.2025.3576008.

45. W. Ning, Y. Zhu, C. Song, et al., "Blockchain-Based Federated Learning: a Survey and New Perspectives," *Applied Sciences* 14, no. 20 (2024): 9459, https://doi.org/10.3390/app14209459.

46. S. M. Rajagopal and R. Buyya, "Blockchain Integrated Federated Learning in edge-fog-cloud Systems for IoT Based Healthcare Applications A Survey," *arXiv [csCR]* (2024).

47. M. Firdaus, H. T. Larasati, and K. Hyune-Rhee, "Blockchain-Based Federated Learning with Homomorphic Encryption for privacy-preserving Healthcare Data Sharing," *Internet of Things* 31 (2025): https://doi.org/10.1016/j.iot.2025.101579.

48. A. Ali, H. Ali, A. Saeed, et al., "Blockchain-Powered Healthcare Systems: Enhancing Scalability and Security with Hybrid Deep Learning," *Sensors* 23, no. 18 (2023): 7740, https://doi.org/10.3390/s23187740.

49. J. Du, J. Hu, Z. Wang, et al., "SoK: on Gradient Leakage in Federated Learning," *arXiv [csCR]* (2025).

50. Y. Li, Z. Guo, N. Yang, H. Chen, D. Yuan, and W. Ding, "Threats and Defenses in the Federated Learning Life Cycle: a Comprehensive Survey and Challenges," *IEEE Transactions on Neural Networks and Learning Systems* 36, no. 9 (2025): 15643–15663, https://doi.org/10.1109/tnnls.2025.3563537.

51. J. Ayeelyan, S. Utomo, A. Rouniyar, H. C. Hsu, and P. A. Hsiung, "Federated Learning Design and Functional Models: Survey," *Artificial Intelligence Review* 58, no. 1 (2025): 21, https://doi.org/10.1007/s10462-024-10969-y.

52. M. Ali, F. Naeem, M. Tariq, and G. Kaddoum, "Federated Learning for Privacy Preservation in Smart Healthcare Systems: a Comprehensive Survey," *IEEE Journal of Biomedical and Health Informatics* 27, no. 2 (2023): 778–789, https://doi.org/10.1109/JBHI.2022.3181823.

53. T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," *arXiv [csLG]* (2020).

54. X. Huang, P. Li, and X. Li, "Stochastic Controlled Averaging for Federated Learning with Communication Compression," *arXiv [mathOC]* (2024).

55. J. Tursunboev, V. Palakonda, Il-M. Kim, S. Moon, and J. Kang, "Multi-Objective Optimisation Framework for Heterogeneous Federated Learning," *CAAI Transactions on Intelligence Technology* 11 (2025): 1–14, https://doi.org/10.1049/cit2.70090.

56. S. Reddi, Z. Charles, M. Zaheer, et al., *Adaptive Federated Optimization* (2021).

57. B. L. Nguyen, T. C. Cao, and B. Le, "Meta-Learning and Personalization Layer in Federated Learning," in *Intelligent Information and Database Systems: 14Th Asian Conference, ACIIDS 2022, Ho Chi Minh City, Vietnam, November 28–30, 2022, Proceedings, Part I* (Springer-Verlag, 2022), 209–221, https://doi.org/10.1007/978-3-031-21743-2_17.

58. S. Ahmed, M. S. Kaiser, S. Chaki, S. Aloteibi, and M. A. Moni, "Federated Learning Model with Dynamic Scoring-based Client Selection for Diabetes Diagnosis," *Knowledge-Based Systems* 320 (2025): https://doi.org/10.1016/j.knosys.2025.113662.

59. X. Li, M. Jiang, X. Zhang, M. Kamp, and Q. Dou, "FedBN: Federated Learning on Non-IID Features via Local Batch Normalization," *arXiv [csLG]* (2021).

60. X. Yang, D. Wan, G. Han, W. Zhang, and W. Tang, "Personalized Federated Learning with Hierarchical Reweighting for multi-center Clinical Prediction," *Computer Methods and Programs in Biomedicine* 271 (2025): https://doi.org/10.1016/j.cmpb.2025.109015.

61. M. Adam and U. Baroudi, "Federated Learning for IoT: Applications, Trends, Taxonomy, Challenges, Current Solutions, and Future Directions," *IEEE Open Journal of the Communications Society* 5 (2024): 7842–7877, https://doi.org/10.1109/OJCOMS.2024.3506214.

62. M. J. Baucas, P. Spachos, and K. N. Plataniotis, "Federated Learning and Blockchain-Enabled Fog-IoT Platform for Wearables in Predictive Healthcare," *IEEE Transactions on Computational Social Systems* 10, no. 4 (2023): 1732–1741, https://doi.org/10.1109/TCSS.2023.3235950.

63. A. Kumar and S. N. Srirama, "FedStrag: Straggler-Aware Federated Learning for Low Resource Devices," *Digital Communications and Networks* 11, no. 4 (2025): 1214–1224, https://doi.org/10.1016/j.dcan.2024.12.004.

64. Y. Shi, J. Nie, X. Li, and H. Li, "Heterogeneity-Aware Device Selection for Efficient Federated Edge Learning," *International Journal of Intelligent Networks* 5 (2024): 293–301, https://doi.org/10.1016/j.ijin.2024.08.002.

65. K. S. Arikumar, S. B. Prathiba, M. Alazab, et al., "FL-PMI: Federated Learning-Based Person Movement Identification Through Wearable Devices in Smart Healthcare Systems," *Sensors* 22, no. 4 (2022): 1377, https://doi.org/10.3390/s22041377.

66. K. S. C and G. Sureshkumar, "Federated Transfer Learning Framework for Multi-disease Prediction," *Procedia Computer Science* 258, no. C (2025): 830–838, https://doi.org/10.1016/j.procs.2025.04.315.

67. Y. Shen, W. Xi, Y. Cai, Y. Fan, H. Yang, and J. Zhao, "Multi-Objective Federated Learning: Balancing Global Performance and Individual Fairness," *Future Generation Computer Systems* 162 (2025): https://doi.org/10.1016/j.future.2024.07.046.

68. W. Zhang, Y. Zhao, F. Li, and H. Zhu, "A Hierarchical Federated Learning Algorithm Based on Time Aggregation in Edge Computing Environment," *Applied Sciences* 13, no. 9 (2023): 5821, https://doi.org/10.3390/app13095821.

69. T. Tang, Z. Han, Z. Cai, et al., "Personalized Federated Graph Learning on Non-IID Electronic Health Records," *IEEE Transactions on Neural Networks and Learning Systems* 35, no. 9 (2024): 11843–11856, https://doi.org/10.1109/TNNLS.2024.3370297.

70. A. Rahman, M. S. Hossain, G. Muhammad, et al., "Federated Learning-based AI Approaches in Smart Healthcare: Concepts, Taxonomies, Challenges and Open Issues," *Cluster Computing* 26, no. 4 (2023): 2271–2311, https://doi.org/10.1007/s10586-022-03658-4.

71. Z. Jia, T. Zhou, Z. Yan, J. Hu, and Y. Shi, "Personalized Meta-Federated Learning for IoT-Enabled Health Monitoring," *Transaction Computation-Aided Design Integration Circuit System* 43, no. 10 (2024): 3157–3170, https://doi.org/10.1109/TCAD.2024.3388908.

72. O. Ozan Güner, "Differentially private fine-tuning of self-supervised Learning Models for Human Activity Recognition on Wearables" (2025).

73. Y. M. Lin, Y. Gao, M. G. Gong, S. J. Zhang, Y. Q. Zhang, and Z. Y. Li, "Federated Learning on Multimodal Data: a Comprehensive Survey," *Machine Intelligent Research* 20, no. 4 (2023): 539–553, https://doi.org/10.1007/s11633-022-1398-0.

74. L. Che, J. Wang, Y. Zhou, and F. Ma, "Multimodal Federated Learning: a Survey," *Sensors* 23, no. 15 (2023): 6986, https://doi.org/10.3390/s23156986.

75. T. Shaik, X. Tao, L. Li, H. Xie, and J. D. Velásquez, "A Survey of Multimodal Information Fusion for Smart Healthcare: Mapping the Journey from Data to Wisdom," *Information Fusion* 102 (2024): https://doi.org/10.1016/j.inffus.2023.102040.

76. J. Thrasher, A. Devkota, P. Siwakotai, et al., "Multimodal Federated Learning in Healthcare: a Review," *arXiv [csLG]* (2024).

77. J. Wu, K. He, R. Mao, X. Shang, and E. Cambria, "Harnessing the Potential of Multimodal EHR Data: a Comprehensive Survey of Clinical Predictive Modeling for Intelligent Healthcare," *Information Fusion* 123 (2025): https://doi.org/10.1016/j.inffus.2025.103283.

78. A. Jabbar, J. Huang, M. K. Jabbar, and A. Ali, "Adaptive Multimodal Fusion in Vertical Federated Learning for Decentralized Glaucoma Screening," *Brain Sciences* 15, no. 9 (2025): 990, https://doi.org/10.3390/brainsci15090990.

79. J. Wang, M. T. Quasim, and B. Yi, "Privacy-Preserving Heterogeneous multi-modal Sensor Data Fusion via Federated Learning for Smart Healthcare," *Information Fusion* 120 (2025): https://doi.org/10.1016/j.inffus.2025.103084.

80. P. Saha, D. Mishra, F. Wagner, K. Kamnitsas, and J. A. Noble, "Incongruent Multimodal Federated Learning for Medical Vision and Language-based multi-label Disease Detection," in *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence and Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence and Fifteenth Symposium on Educational Advances in Artificial Intelligence*, 39, no. 27 (April 2025), 28331–28339, https://doi.org/10.1609/aaai.v39i27.35054.

81. A. Orzikulova, J. Kwak, J. Shin, and S. J. Lee, "Federated Learning for Time-Series Healthcare Sensing with Incomplete Modalities," *arXiv [csLG]* (2024).

82. S. U. Khan, M. Sultana, S. Danish, et al., "Multimodal Feature Fusion for Human Activity Recognition Using Human Centric Temporal Transformer," *Engineering Applications of Artificial Intelligence* 160 (2025): https://doi.org/10.1016/j.engappai.2025.111844.

83. S. M. Sheikholeslami, P. C. Ng, J. Abouei, and K. N. Plataniotis, "Multi-Modal Federated Learning over Cell-Free Massive MIMO Systems for Activity Recognition," *IEEE Access* 13 (2025): 40844–40858, https://doi.org/10.1109/ACCESS.2025.3548001.

84. T. Muazu, Y. Mao, A. U. Muhammad, M. Ibrahim, U. M. M. Kumshe, and O. Samuel, "A Federated Learning System with Data Fusion for Healthcare Using multi-party Computation and Additive Secret Sharing," *Computer Communications* 216 (2024): 168–182, https://doi.org/10.1016/j.comcom.2024.01.006.

85. X. Chen, H. Xie, X. Tao, F. L. Wang, M. Leng, and B. Lei, "Artificial Intelligence and Multimodal Data Fusion for Smart Healthcare: Topic Modeling and Bibliometrics," *Artificial Intelligence Review* 57, no. 4 (2024): 91, https://doi.org/10.1007/s10462-024-10712-7.

86. Y. Wang, "Multimodal Federated Learning for Healthcare: Adaptive and Private Fusion," *Applied and Computational Engineering* 166, no. 1 (2025): 94–101, https://doi.org/10.54254/2755-2721/2025.TJ24497.

87. D. N. Sachin, B. Annappa, S. Ambasange, and A. E. Tony, "A Multimodal Contrastive Federated Learning for Digital Healthcare," *SN Computer Science* 4, no. 5 (2023): https://doi.org/10.1007/s42979-023-02124-1.

88. L. Zong, Q. Xie, J. Zhou, P. Wu, X. Zhang, and B. Xu, "FedCMR: Federated Cross-Modal Retrieval," in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval* (April 2021), 1672–1676, https://doi.org/10.1145/3404835.3462989.

89. T. Li, Y. Gong, H. Liu, Z. Chen, and L. Song, "Communication-Efficient Multi-Modal Federated Learning via Dynamic Client-Modality Matching," in *2024 IEEE International Conference on Big Data (BigData)* (Washington, DC, USA, June 2024), 7899–7908, https://doi.org/10.1109/BigData62323.2024.10825371.

90. T. Elfouly and A. Alouani, "A Comprehensive Survey on Wearable Computing for Mental and Physical Health Monitoring," *Electronics* 14, no. 17 (2025): 3443, https://doi.org/10.3390/electronics14173443.

91. X. Chen, W. Wu, F. Ji, Y. Lu, and L. Li, "Privacy-Aware Split Federated Learning for LLM Fine-Tuning over Internet of Things," *IEEE Internet of Things Journal* 12, no. 24 (2025): 51902–51913, https://doi.org/10.1109/JIOT.2025.3600269.

92. Y. Shi, M. Xue, Y. Zeng, J. Zhang, J. Wan, and Y. Zhou, "FedAMM: Federated Learning for Brain Tumor Segmentation with Arbitrary Missing Modalities," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025: 28Th International Conference, Daejeon, South Korea, September 23–27, 2025, Proceedings, Part VIII* (Springer-Verlag, 2025), 203–213, https://doi.org/10.1007/978-3-032-04984-1_20.

93. A. A. Abdellatif, N. Mhaisen, A. Mohamed, et al., "Communication-Efficient Hierarchical Federated Learning for IoT Heterogeneous Systems with Imbalanced Data," *Future Generation Computer Systems* 128 (2022): 406–419, https://doi.org/10.1016/j.future.2021.10.016.

94. Y. He, D. Yan, and F. Chen, "Hierarchical Federated Learning with Local Model Embedding," *Engineering Applications of Artificial Intelligence* 123 (2023): https://doi.org/10.1016/j.engappai.2023.106148.

95. C. Ma, X. Li, B. Huang, G. Li, and F. Li, "Personalized client-edge-cloud Hierarchical Federated Learning in Mobile Edge Computing," *Journal of Cloud Computing* 13, no. 1 (2024): 161, https://doi.org/10.1186/s13677-024-00721-w.

96. L. Song, J. Li, H. Jiang, S. Wei, and Y. Guo, "CHPFL: Clustered Adaptive Hierarchical Federated Learning for edge-level Personalization," *High-Confidence Computing* 6, no. 1 (2025): https://doi.org/10.1016/j.hcc.2025.100343.

97. N. M. Eshwarappa, H. Baghban, C. H. Hsu, P. Y. Hsu, R. H. Hwang, and M. Y. Chen, "Communication-Efficient and privacy-preserving Federated Learning for Medical Image Classification in multi-institutional Edge Computing," *Journal of Cloud Computing* 14, no. 1 (2025): https://doi.org/10.1186/s13677-025-00734-z.

98. C. Xu, Y. Qu, Y. Xiang, and L. Gao, "Asynchronous Federated Learning on Heterogeneous Devices: a Survey," *Computer Science Review* 50 (2023): https://doi.org/10.1016/j.cosrev.2023.100595.

99. B. Fan, X. Su, S. Tarkoma, and P. Hui, "Behave Differently when Clustering: a Semi-asynchronous Federated Learning Approach for IoT," *ACM Trans Sen Netw* 20, no. 3 (2024): 1–28, https://doi.org/10.1145/3639825.

100. J. Li, T. Chen, and S. Teng, "A Comprehensive Survey on Client Selection Strategies in Federated Learning," *Computer Networks* 251 (2024): https://doi.org/10.1016/j.comnet.2024.110663.

101. C. Messinis S, *E.* Protonotarios N, E. Arapidis, and N. Doulamis, "Client Selection and Resource Allocation via Graph Neural Networks for Efficient Federated Learning in Healthcare Environments," in *Proceedings of the 17th International Conference on PErvasive Technologies Related to Assistive Environments. PETRA'24* (Association for Computing Machinery, June 2024), 606–612, https://doi.org/10.1145/3652037.3663906.

102. K. Wei, J. Li, C. Ma, et al., "Gradient Sparsification for Efficient Wireless Federated Learning with Differential Privacy," *Science China Information Sciences* 67, no. 4 (2024): https://doi.org/10.1007/s11432-023-3918-9.

103. C. Yu, S. Shen, S. Wang, K. Zhang, and H. Zhao, "Communication-Efficient Hybrid Federated Learning for E-Health with Horizontal and Vertical Data Partitioning," *IEEE Transactions on Neural Networks and Learning Systems* 36, no. 3 (2025): 5614–5628, https://doi.org/10.1109/TNNLS.2024.3383748.

104. S. Alahmari and I. Alghamdi, "A Comprehensive Survey on energy-efficient and privacy-preserving Federated Learning for Edge Intelligence and IoT," *Results in Engineering* 28 (2025): https://doi.org/10.1016/j.rineng.2025.107849.

105. Z. Dakhia and M. Merenda, “Client Selection in Federated Learning on Resource-Constrained Devices: a Game Theory Approach,” *Applied Sciences* 15, no. 13 (2025): 7556, https://doi.org/10.3390/app15137556.

106. H. Hilberger, S. Hanke, and M. Bödenler, “Federated Learning with Dynamic Model Exchange,” *Electronics* 11, no. 10 (2022): 1530, https://doi.org/10.3390/electronics11101530.

107. L. Wu, Y. Jin, and K. Hao, “Optimized Compressed Sensing for Communication Efficient Federated Learning,” *Knowledge-Based Systems* 278 (2023): https://doi.org/10.1016/j.knosys.2023.110805.

108. Y. Ren, Y. Cao, C. Ye, and X. Cheng, “Two-Layer Accumulated Quantized Compression for communication-efficient Federated Learning: TLAQC,” *Scientific Reports* 13, no. 1 (2023): https://doi.org/10.1038/s41598-023-38916-x.

109. Z. He, G. Zhu, S. Zhang, E. Luo, and Y. Zhao, “FedDT: a Communication-Efficient Federated Learning via Knowledge Distillation and Ternary Compression,” *Electronics* 14, no. 11 (2025): 2183, https://doi.org/10.3390/electronics14112183.

110. M. Chen, N. Shlezinger, H. V. Poor, Y. C. Eldar, and S. Cui, “Communication-Efficient Federated Learning,” *Proceedings of the National Academy of Sciences* 118, no. 17 (2021): https://doi.org/10.1073/pnas.2024789118.

111. S. Cao, H. Zhang, T. Wen, et al., “FedQMIX: Communication-Efficient Federated Learning via multi-agent Reinforcement Learning,” *High-Confidence Computing* 4, no. 2 (2024): https://doi.org/10.1016/j.hcc.2023.100179.

112. E. Dritsas and M. Trigka, “Federated Learning for IoT: a Survey of Techniques, Challenges, and Applications,” *Journal of Sensor and Actuator Networks* 14, no. 1 (2025): 9, https://doi.org/10.3390/jsan14010009.

113. S. Mohammadi, A. Balador, S. Sinaei, and F. Flammini, “Balancing Privacy and Performance in Federated Learning: a Systematic Literature Review on Methods and Metrics,” *Journal of Parallel and Distributed Computing* 192 (2024): https://doi.org/10.1016/j.jpdc.2024.104918.

114. S. Chen, W. Liu, X. Zhang, H. Xu, W. Lin, and X. Chen, “Adaptive Personalized Federated Learning for Non-IID Data with Continual Distribution Shift,” *2024 IEEE/ACM 32nd International Symposium on Quality of Service (IWQoS)* (Guangzhou, China, 2024), 1–6, https://doi.org/10.1109/IWQoS61813.2024.10682851.

115. A. Choudhury, K. K. Sarma, V. Gulvanskii, D. Kaplun, and L. Dutta, “Leveraging Federated Learning and Edge Computing for pandemic-resilient Healthcare,” *Scientific Reports* 15, no. 1 (2025): https://doi.org/10.1038/s41598-025-00199-9.

116. S. Alzu’bi, T. Kanan, M. Elbes, G. Kanaan, and I. Trrad, “Energy-Efficient Edge Deployment of Generative AI Models Using Federated Learning,” *Cluster Computing* 28, no. 5 (2025): 315, https://doi.org/10.1007/s10586-025-05263-7.

117. M. E. Yahiaoui, M. Derdour, R. Abdulghafor, et al., “Federated Learning with Privacy Preserving for Multi- Institutional Three-Dimensional Brain Tumor Segmentation,” *Diagnostics* 14, no. 24 (2024): 2891, https://doi.org/10.3390/diagnostics14242891.

118. X. Zhang, H. Deng, R. Wu, J. Ren, and Y. Ren, “PQSF: Post-Quantum Secure privacy-preserving Federated Learning,” *Scientific Reports* 14, no. 1 (2024): https://doi.org/10.1038/s41598-024-74377-6.

119. D. C. Nguyen, M. R. Uddin, S. Shaon, R. Rahman, O. Dobre, and D. Niyato, “Quantum Federated Learning: a Comprehensive Survey,” *arXiv [csLG]* (2025).

120. R. Ballester, J. Cerquides, and L. Artiles, “Quantum Federated Learning: a Comprehensive Literature Review of Foundations, Challenges, and Future Directions,” *Quantum Machine Intelligent* 7, no. 2 (2025): 73, https://doi.org/10.1007/s42484-025-00292-2.

121. X.-T. Dang, B.-M. Vu, Q.-S. Nguyen, T.-T.-M. Tran, J.-S. Eom, and O.-S. Shin, “A Survey on Energy-Efficient Design for Federated Learning over Wireless Networks,” *Energies* 17, no. 24 (2024): 6485, https://doi.org/10.3390/en17246485.

122. R. Albelaihi, L. Yu, W. D. Craft, X. Sun, C. Wang, and R. Gazda, “Green Federated Learning via Energy-Aware Client Selection,” in *GLOBECOM 2022-2022 IEEE Global Communications Conference* (Rio de Janeiro, Brazil, August 2022), 13–18, https://doi.org/10.1109/GLOBECOM48099.2022.10001569.

123. B. R. D. Guru, “Integrated Data and AI Governance Framework: a Lifecycle Approach to Responsible AI Implementation,” *Journal of Computer Science and Technology Studies* 7, no. 7 (2025): 771–777, https://doi.org/10.32996/jcsts.2025.7.7.83.

124. W. Moulahi, I. Jdey, T. Moulahi, M. Alawida, and A. Alabdulatif, “A Blockchain-based Federated Learning Mechanism for Privacy Preservation of Healthcare IoT Data,” *Computers in Biology and Medicine* 167 (2023): https://doi.org/10.1016/j.compbiomed.2023.107630.